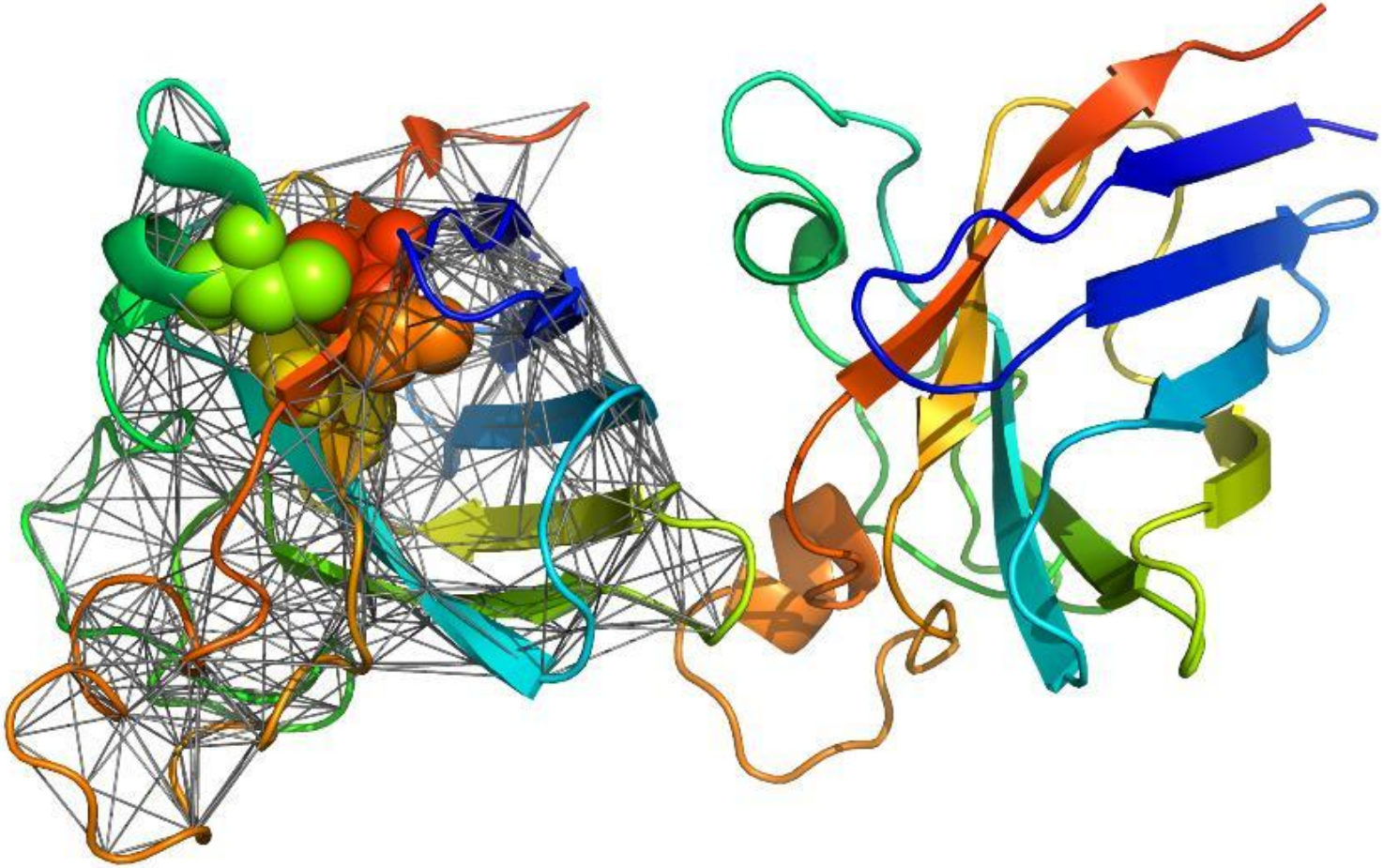
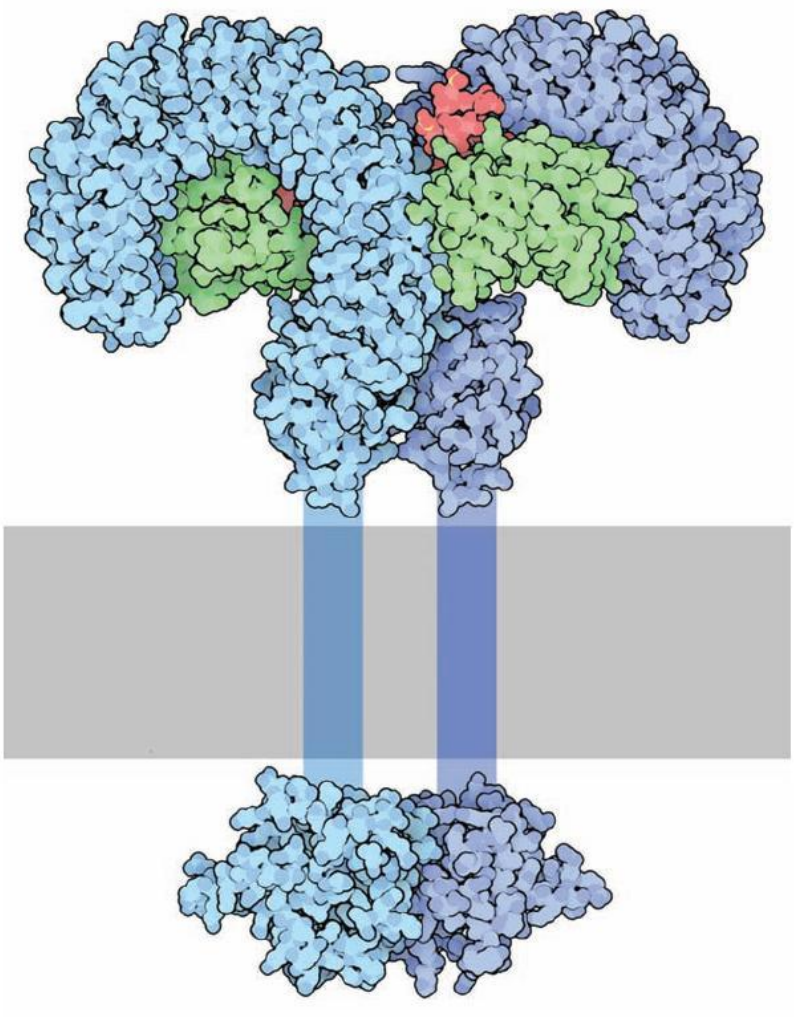
Modeling Protein Structures as Networks



TOLL-LIKE RECEPTORS

frontline for the immune system



3fxi (B.S. Park, et al. *Nature 458: 1191-1195, 2009)*

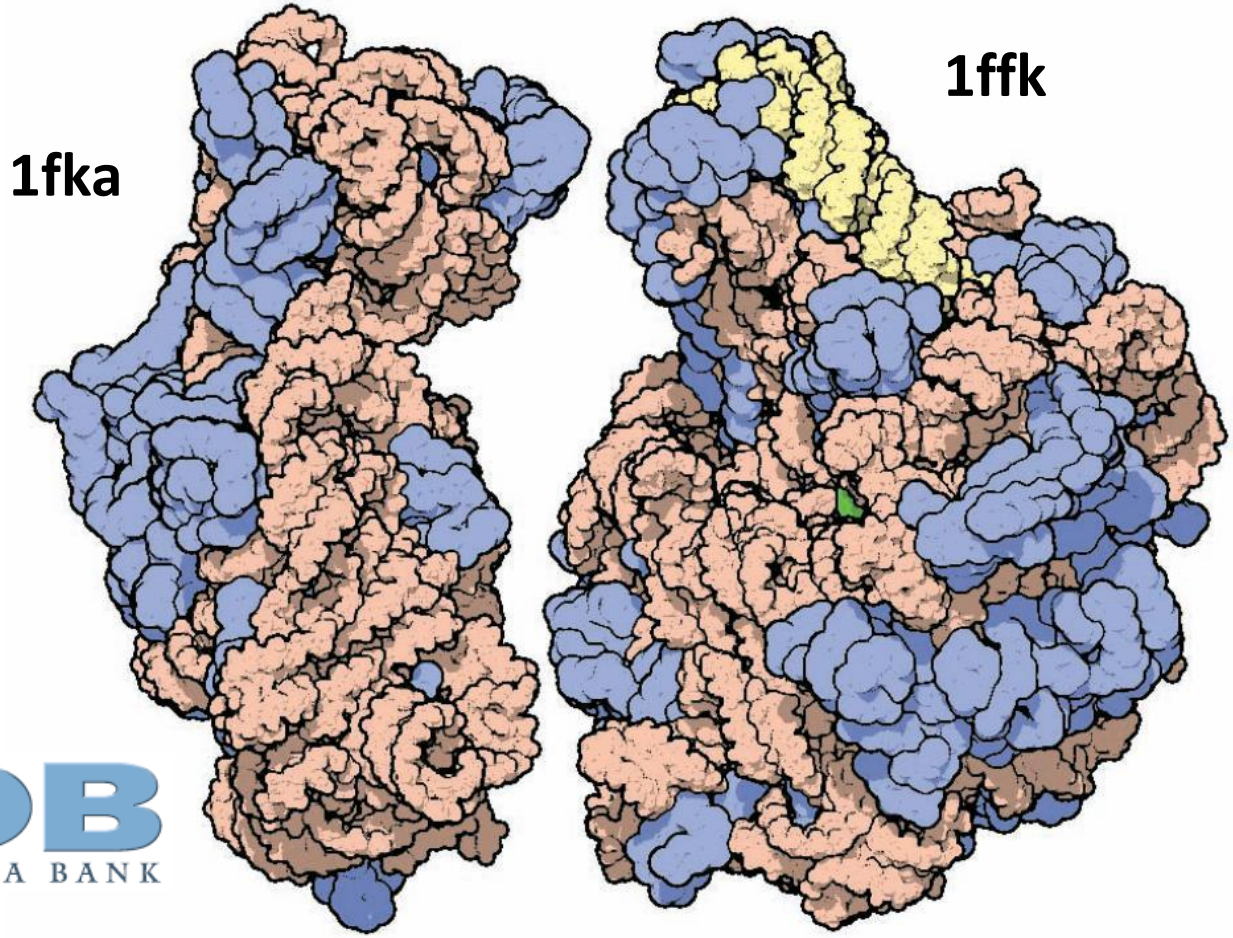
2j67 *(T. Nyman et al., J.Biol.Chem. 283: 11861, 2008).*

RCSB PDB PROTEIN DATA BANK

The 2011 Nobel Prize in Physiology or Medicine: jointly to Bruce A. Beutler (¼) and Jules A. Hoffmann (¼) and Ralph M. Steinman (½)

RIBOSOME

the factory of protein synthesis

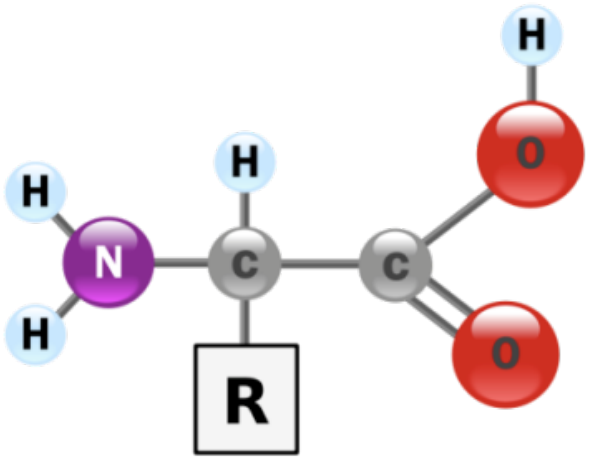


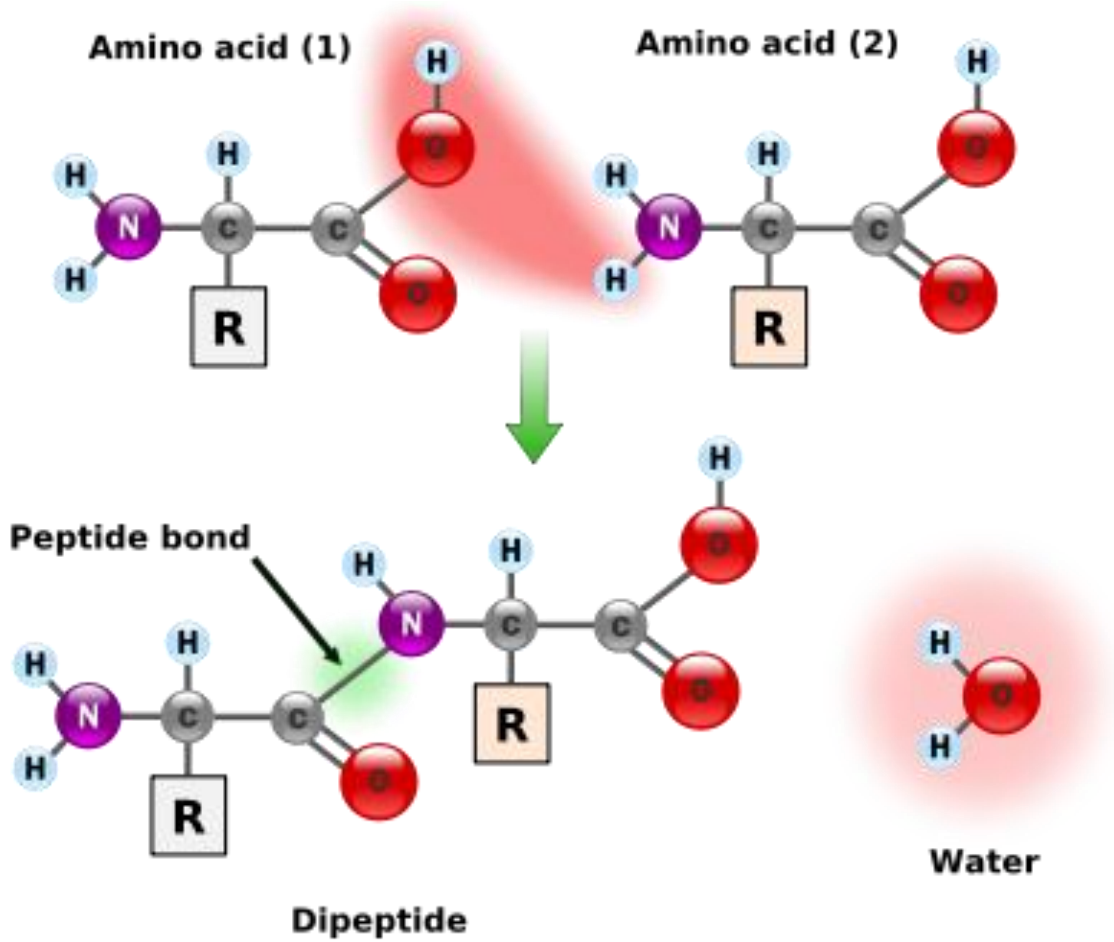
RCSB PDB
PROTEIN DATA BANK

The 2009 Nobel Prize in Chemistry: for studies of the structure and function of the ribosome–Venkatraman Ramakrishnan, Thomas A. Steitz, and Ada E. Yonath.

Proteins are polymers

Proteins are linear polypeptide chains of amino acids.

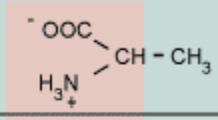




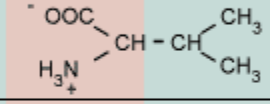
R GROUPS

NONPOLAR, HYDROPHOBIC

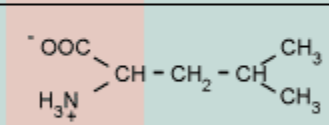
Alanine
Ala
A
MW = 89



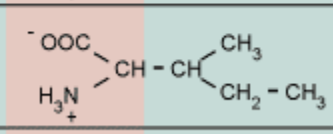
Valine
Val
V
MW = 117



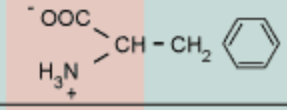
Leucine
Leu
L
MW = 131



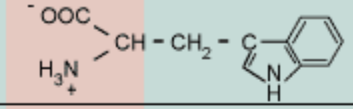
Isoleucine
Ile
I
MW = 131



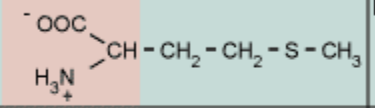
Phenylalanine
Phe
F
MW = 131



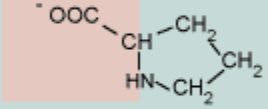
Tryptophan
Trp
W
MW = 204



Methionine
Met
M
MW = 149

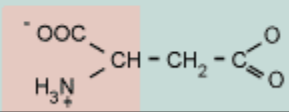


Proline
Pro
P
MW = 115

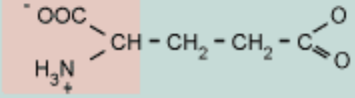


POLAR ACIDIC

Aspartic acid
Asp
D
MW = 133

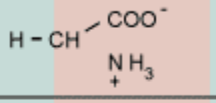


Glutamine acid
Glu
E
MW = 147

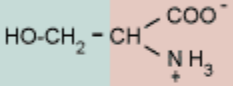


POLAR, UNCHARGED

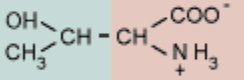
Glycine
Gly
G
MW = 75



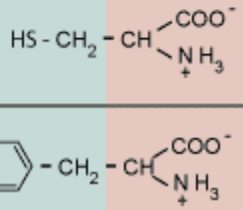
Serine
Ser
S
MW = 105



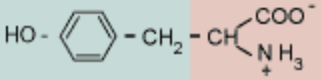
Threonine
Thr
T
MW = 119



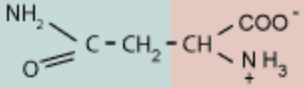
Cysteine
Cys
C
MW = 121



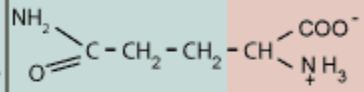
Tyrosine
Tyr
Y
MW = 181



Asparagine
Asn
N
MW = 132

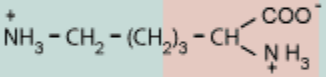


Glutamine
Gln
Q
MW = 146

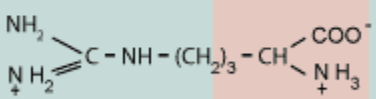


POLAR BASIC

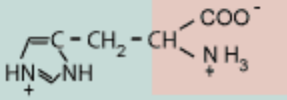
Lysine
Lys
K
MW = 146



Arginine
Arg
R
MW = 174

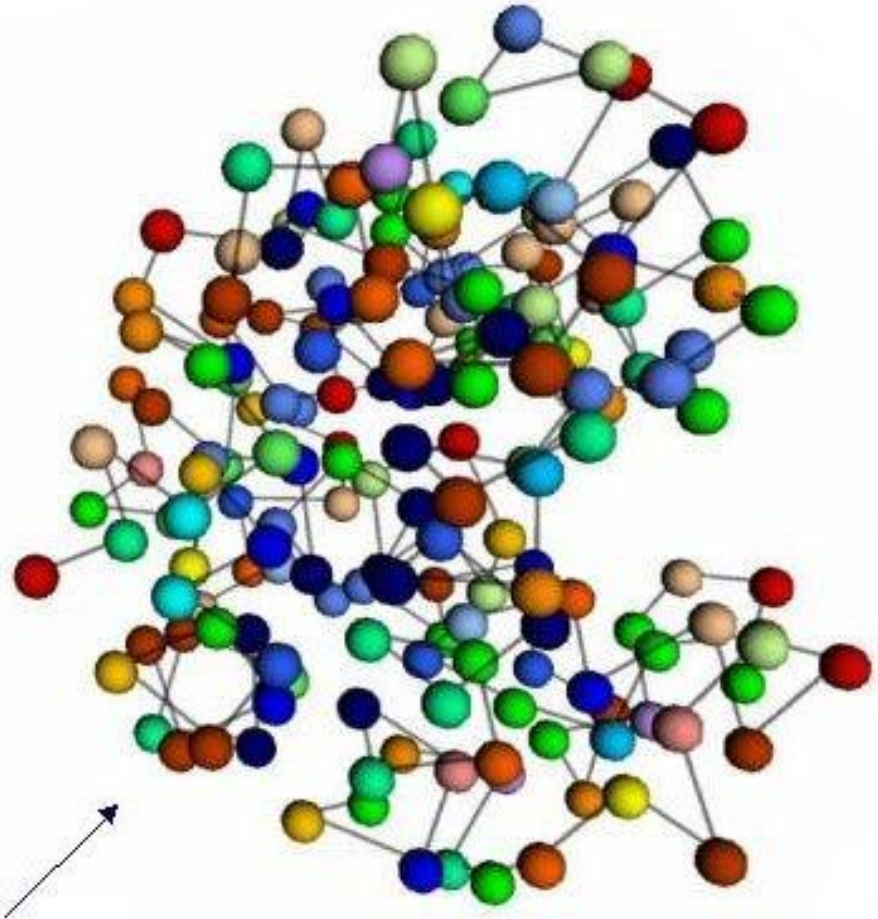


Histidine
His
H
MW = 155



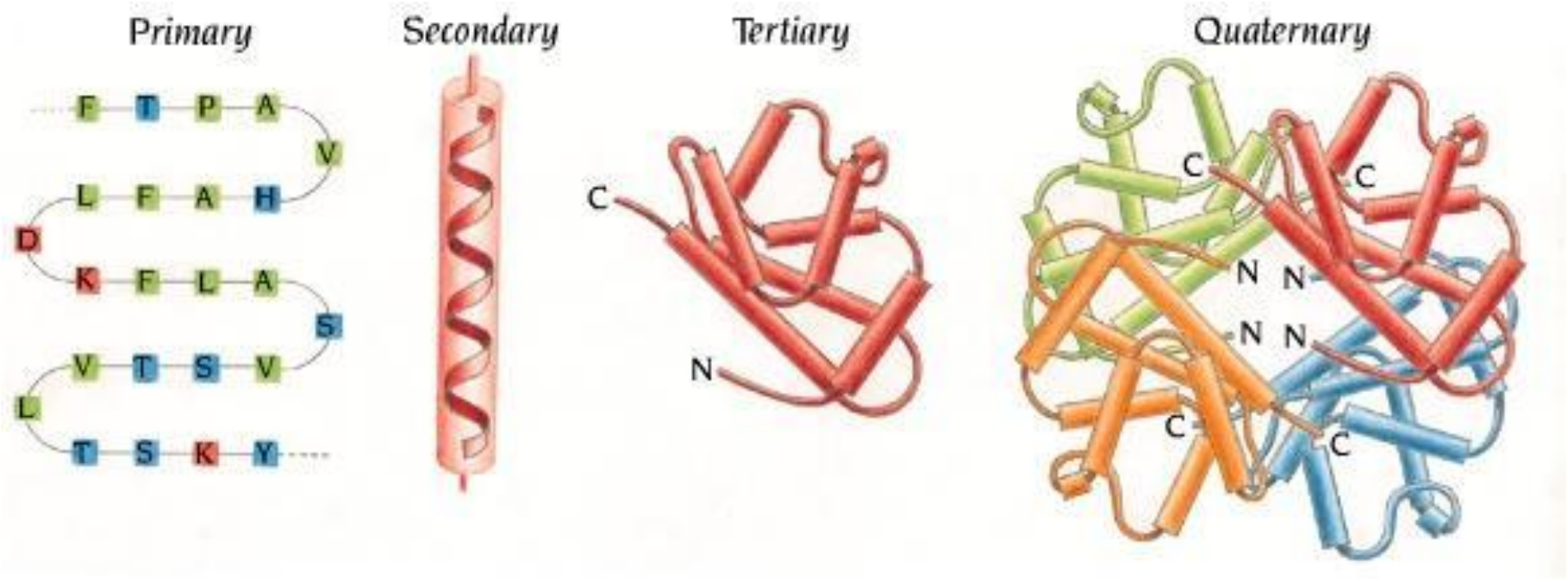
Polypeptide chains fold into 3D protein structures

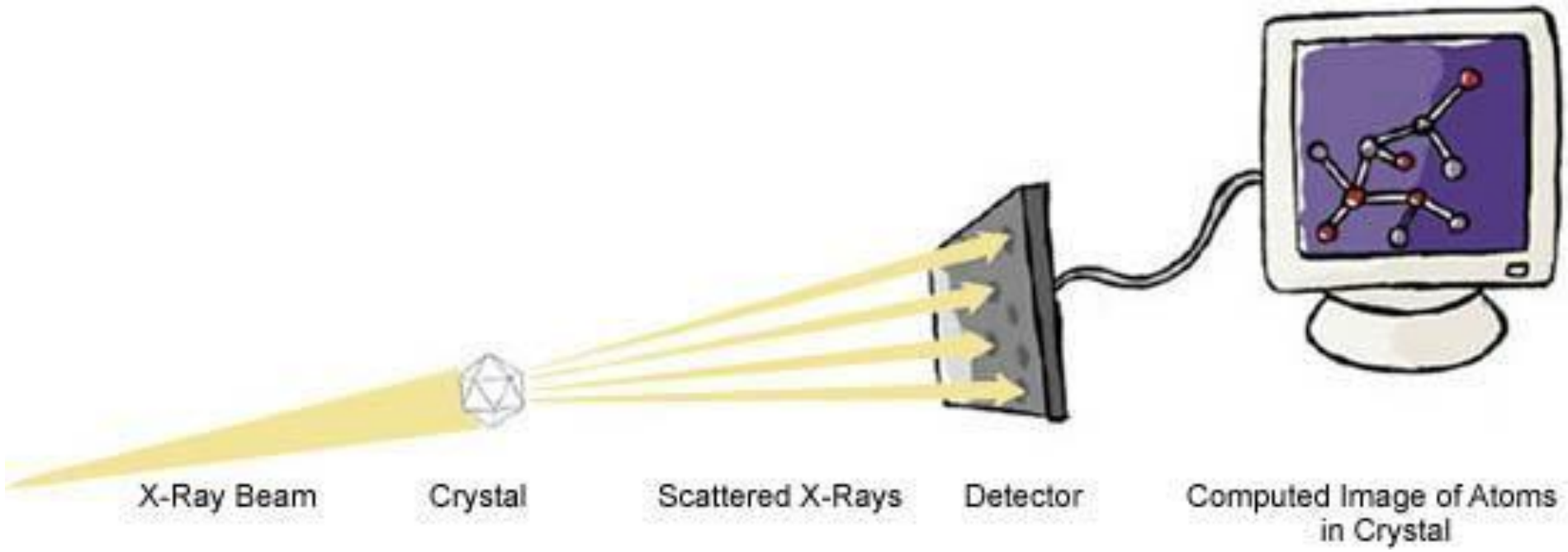
R T D C Y G N V N R I D T T G
A S C K T A K P E G L S Y C G
V S A S K K I A E R D L Q A M
D R Y K T I I K K V G E K L C
V E P A V I A G I I S R E S H
A G K V L K N G W G D R G N G
F G L M Q V D K R S H K P Q G
T W N G E V H I T Q G T T I L
I N F I K T I Q K K F P S W T
K D Q Q L K G G I S A Y N A G
A G N V R S Y A R M D I G T T
H D D Y A N D V V A R A Q Y Y
K Q H G Y



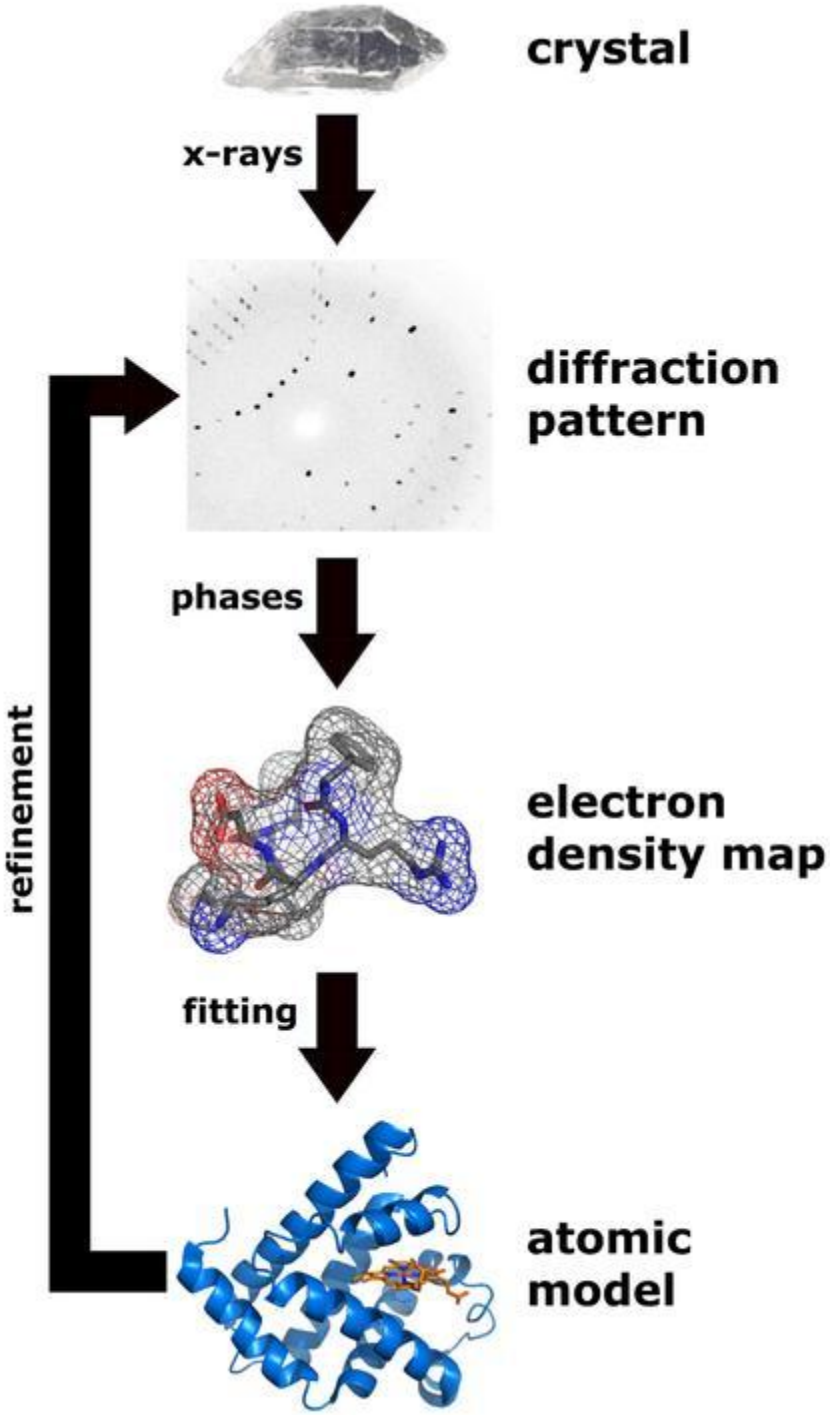
Primary Sequence 3D Structure

Hierarchy in protein structure organization

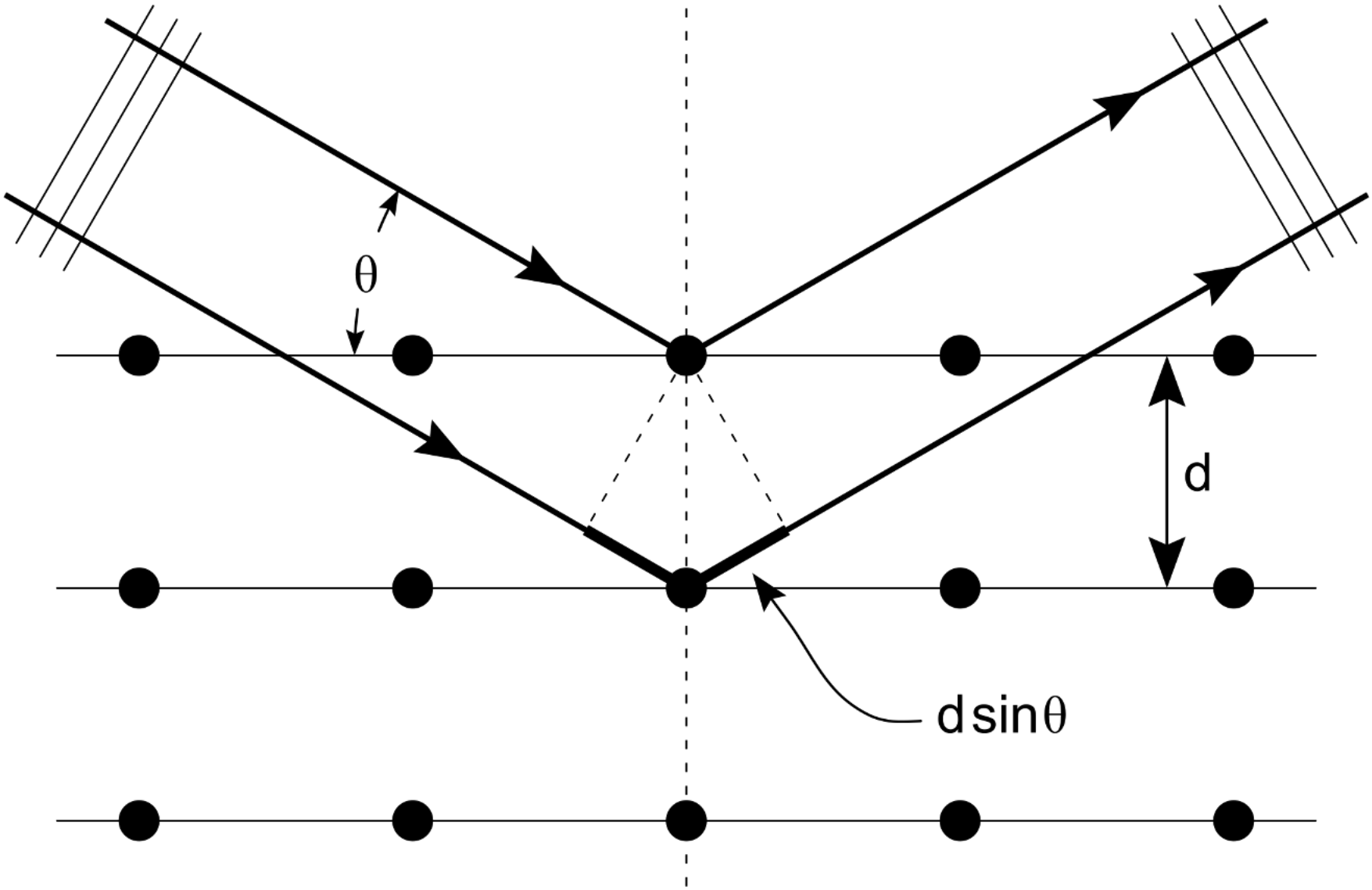


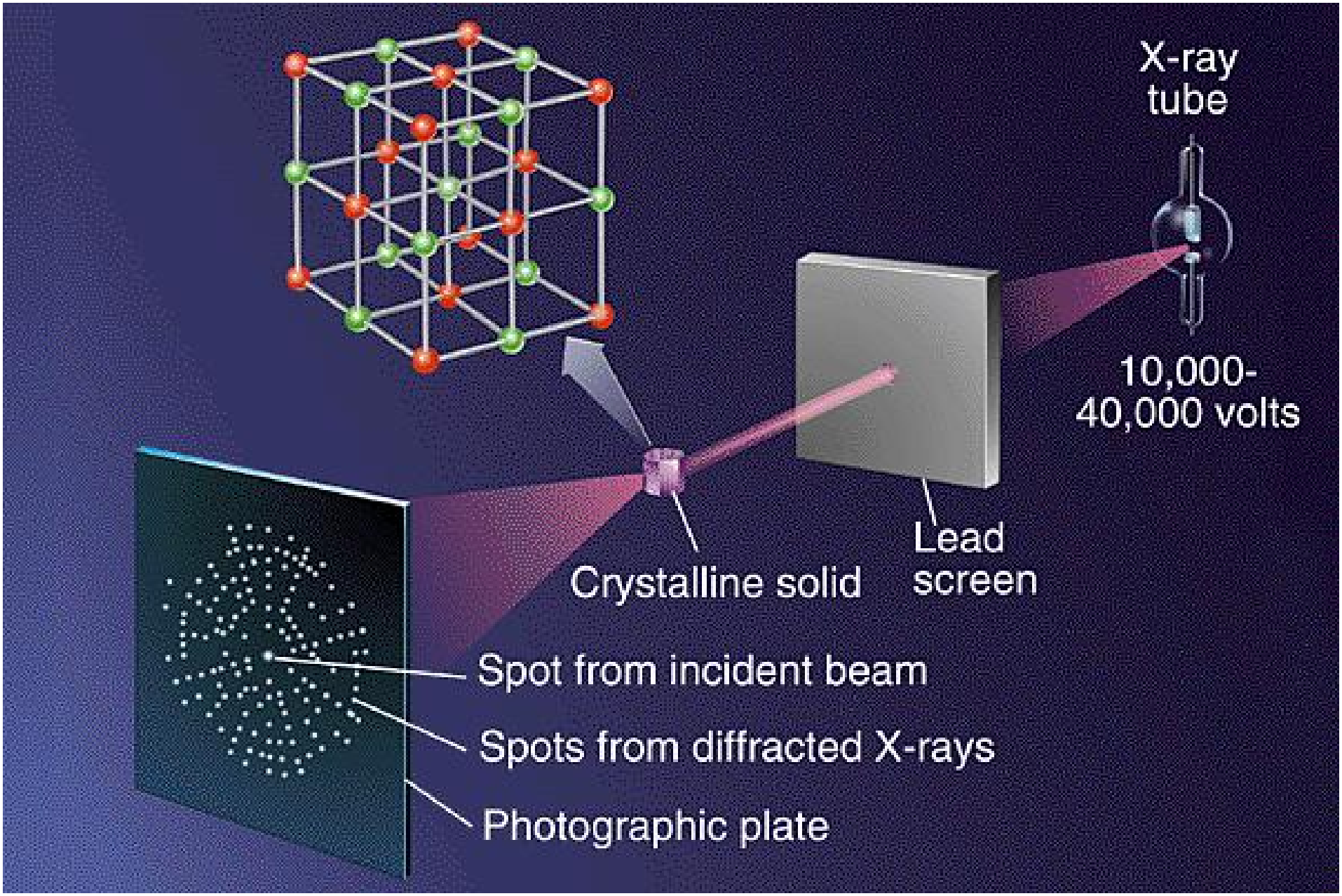


http://publications.nigms.nih.gov/structlife/images/ch2_xray.jpg

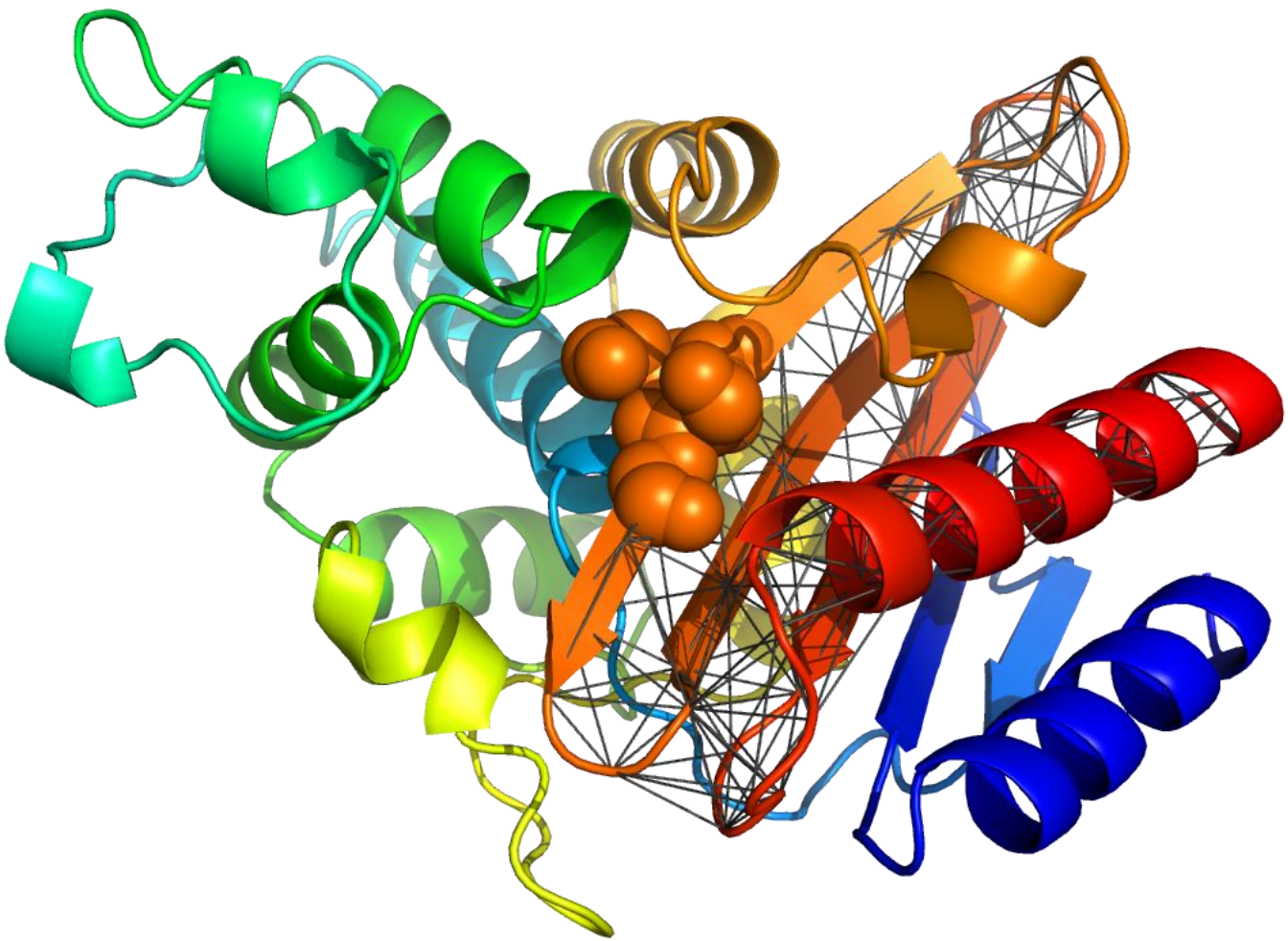

crystal
x-rays
diffraction pattern
phases
electron density map
fitting
atomic model
refinement

Bragg's Law




X-ray
tube
10,000-
40,000 volts
Lead
screen
Crystalline solid
Spot from incident beam
Spots from diffracted X-rays
Photographic plate

RESIDUE INTERACTION GRAPHS



b-Lactamase class A family

Protein residue networks

14

Chapter 14, The Structure of Complex Networks, Ernesto Estrada, OUP.

RESIDUE INTERACTION GRAPHS (RIG & LIN)

❖ **Residue Interaction Graphs (RIG)** is a coarse-grained model of native-state protein structure, based on spatial **contacts** made by residues in the polypeptide chain.

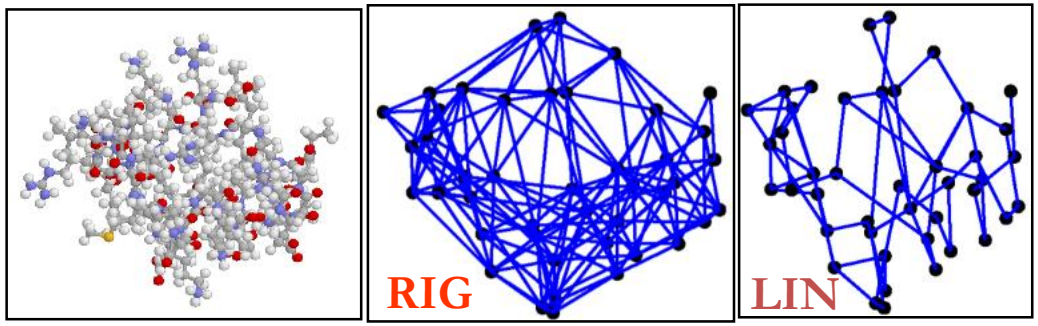
— Any two residues of a protein are said to be in spatial **contact**, if they are within a threshold distance; **Rc ≤ 8 Å**.

(Results are independent of Rc, for 5 Å ≤ Rc ≤ 10 Å.)

❖ **Long-range Interaction Network (LIN)** network is a subset of RIG in which only those spatial **contacts** are considered which are made between residues which are **distant** along the polypeptide chain.

— Any two residues are said to be **distant** if there are a threshold number of residues separating them. $\mathbf{LRI_{Threshold} = 12}$.

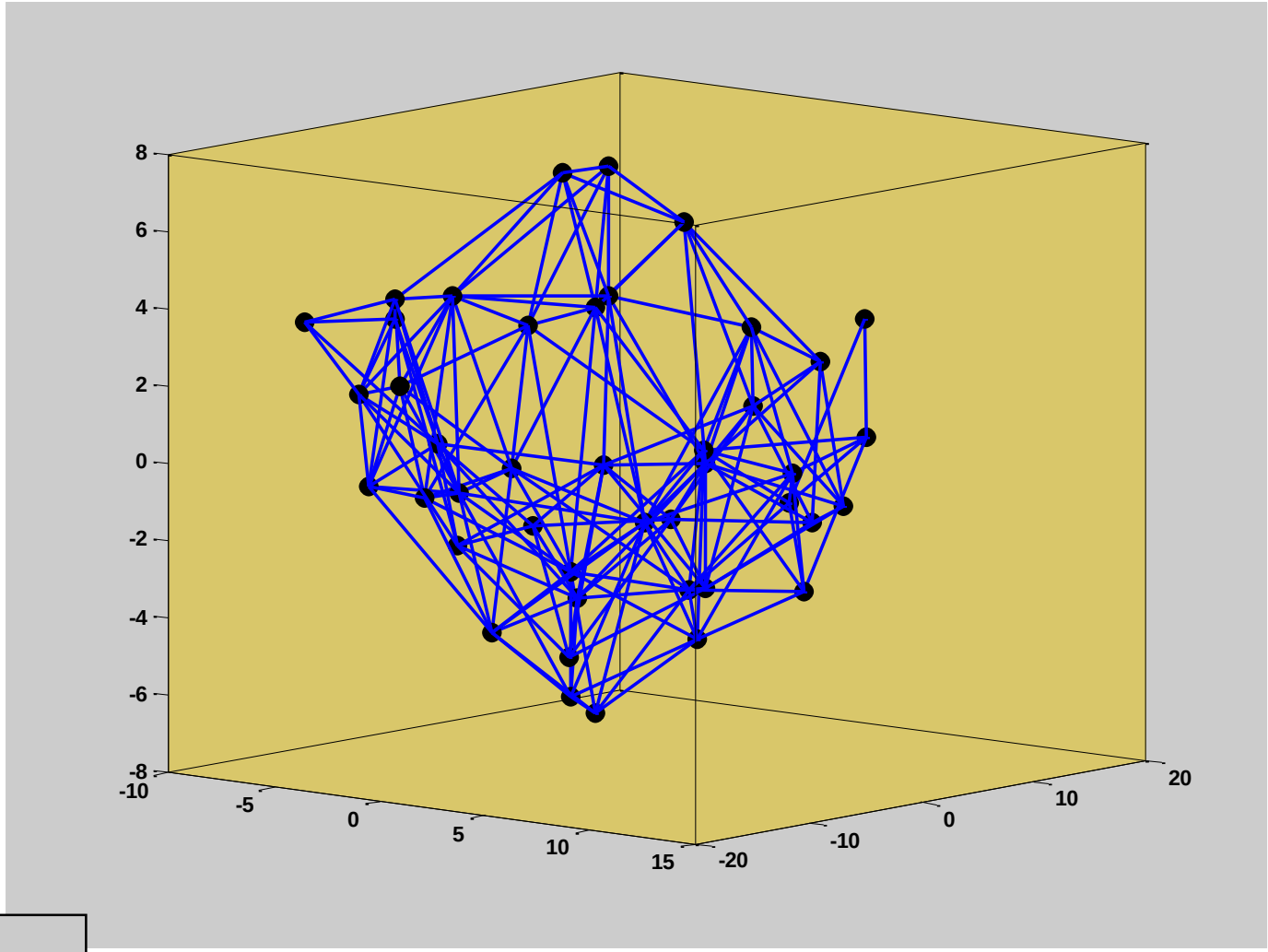
(Typical length of an α helix and a β strand are 11 and 6 residues, respectively.*)

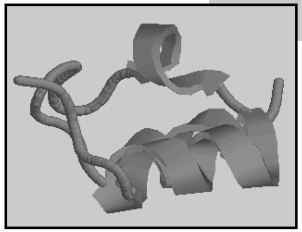


* Schultz and Schirmer, *Princ. Of Prot. Struct.*, Springer (NY),66—107, (1979).

BUILDING RESIDUE INTERACTION GRAPH MODELS

Illustration of Protein Contact Network (PCN) for a small protein, Acyltransferase.

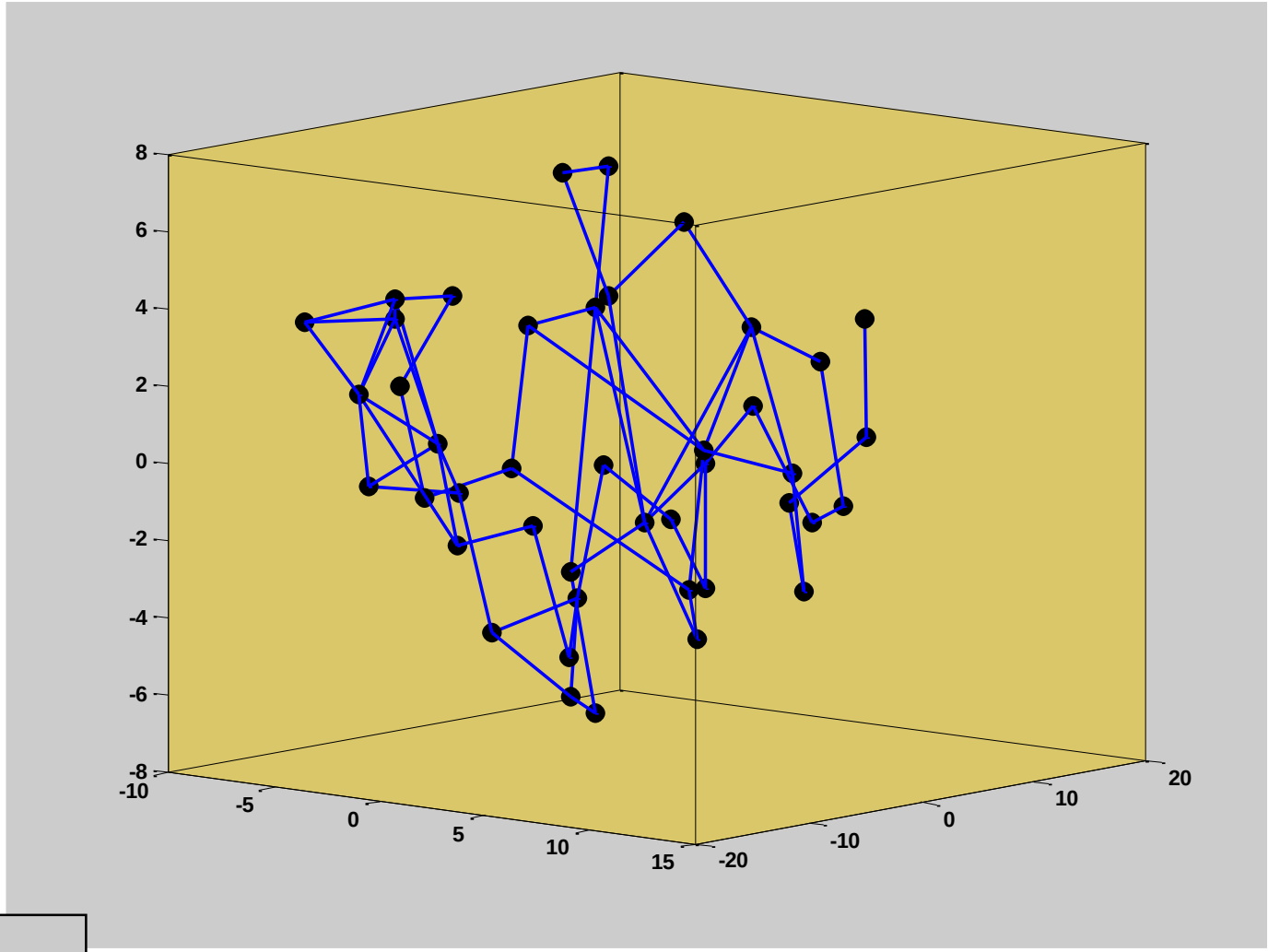


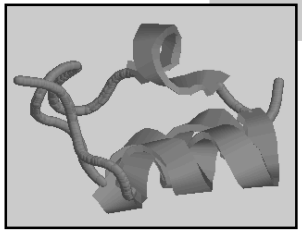


Residue Interaction Graphs (RIG)

BUILDING RESIDUE INTERACTION GRAPH MODELS

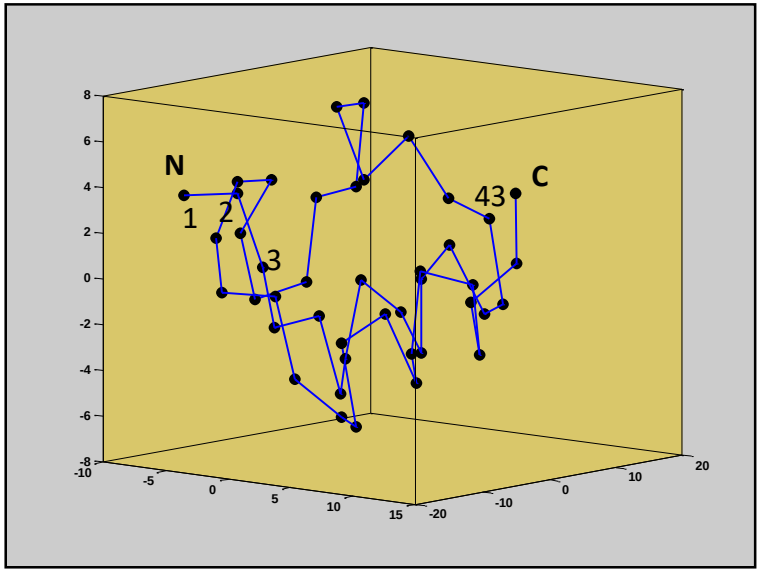
Long-range interactions/contacts in Acyltransferase.

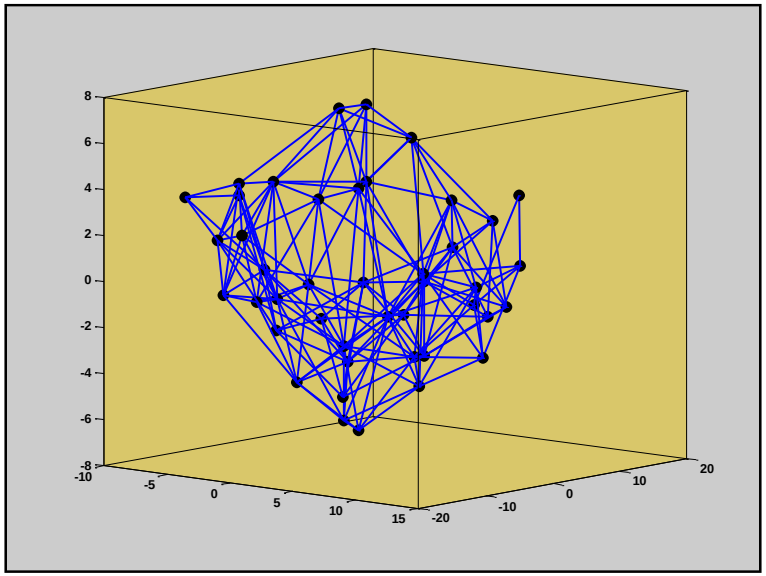


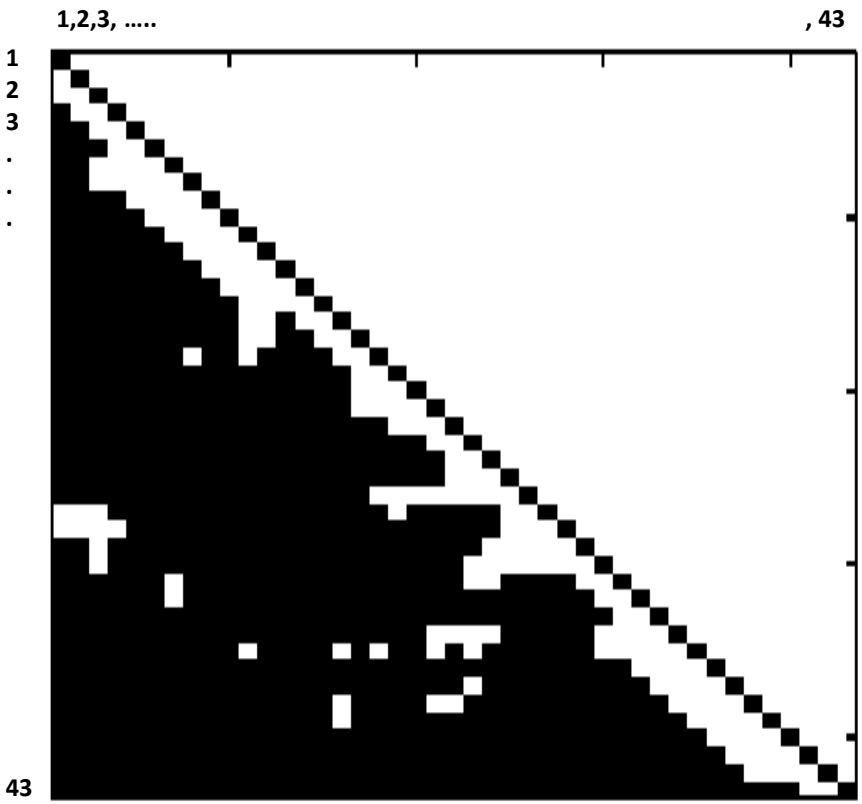


Long-range Interaction Network (LIN)

RESIDUE INTERACTION GRAPH (RIG): A course-grained model







Residues *not* in contact.

Residues make a contact.

Protein backbone with secondary structure traces

Fig. 14.2
Protein backbone. Representation of the three-dimensional structure of the backbone of a folded protein.

The PDB File

```
HEADER    OXYGEN STORAGE                          06-JAN-95   1ASH
TITLE     THE STRUCTURE OF ASCARIS HEMOGLOBIN DOMAIN I AT 2.2
TITLE    2 ANGSTROMS RESOLUTION: MOLECULAR FEATURES OF OXYGEN AVIDITY
COMPND    MOL_ID: 1;
COMPND   2 MOLECULE: HEMOGLOBIN (OXY);
COMPND   3 CHAIN: A;
COMPND   4 ENGINEERED: YES
SOURCE    MOL_ID: 1;
SOURCE   2 ORGANISM_SCIENTIFIC: ASCARIS SUUM;
SOURCE   3 ORGANISM_COMMON: PIG ROUNDWORM;
SOURCE   4 ORGANISM_TAXID: 6253;
SOURCE   5 EXPRESSION_SYSTEM_PLASMID: PET-8C
KEYWDS    OXYGEN STORAGE
EXPDTA    X-RAY DIFFRACTION
AUTHOR    J.YANG,F.S.MATHEWS,A.P.KLOEK,D.E.GOLDBERG
REVDAT   2   24-FEB-09 1ASH    1       VERSN
REVDAT   1   27-FEB-95 1ASH    0
JRNL        AUTH   J.YANG,A.P.KLOEK,D.E.GOLDBERG,F.S.MATHEWS
JRNL        TITL   THE STRUCTURE OF ASCARIS HEMOGLOBIN DOMAIN I AT
JRNL        TITL 2 2.2 A RESOLUTION: MOLECULAR FEATURES OF OXYGEN
JRNL        TITL 3 AVIDITY.
JRNL        REF    PROC.NATL.ACAD.SCI.USA        V.  92  4224 1995
JRNL        REFN                   ISSN 0027-8424
JRNL        PMID   7753786
JRNL        DOI    10.1073/PNAS.92.10.4224
```

The PDB File

ATOM	1	N	ALA	A	0	11.081	1.847	9.557	1.00	64.40	N
ATOM	2	CA	ALA	A	0	10.369	0.997	10.519	1.00	60.28	C
ATOM	3	C	ALA	A	0	9.181	0.318	9.833	1.00	56.39	C
ATOM	4	O	ALA	A	0	9.344	-0.374	8.829	1.00	52.67	O
ATOM	5	N	ASN	A	1	7.992	0.669	10.335	1.00	55.08	N
ATOM	6	CA	ASN	A	1	6.691	0.239	9.830	1.00	53.00	C
ATOM	7	C	ASN	A	1	6.406	0.688	8.391	1.00	47.79	C
ATOM	8	O	ASN	A	1	5.757	-0.071	7.678	1.00	46.55	O
ATOM	9	CB	ASN	A	1	5.615	0.752	10.808	1.00	60.06	C
ATOM	10	CG	ASN	A	1	4.159	0.416	10.450	1.00	65.81	C
ATOM	11	OD1	ASN	A	1	3.751	-0.731	10.250	1.00	68.33	O
ATOM	12	ND2	ASN	A	1	3.303	1.433	10.367	1.00	68.59	N
ATOM	13	N	LYS	A	2	6.837	1.833	7.838	1.00	43.12	N
ATOM	14	CA	LYS	A	2	6.677	1.983	6.389	1.00	38.31	C
ATOM	15	C	LYS	A	2	7.612	1.040	5.632	1.00	31.24	C
ATOM	16	O	LYS	A	2	7.173	0.454	4.658	1.00	35.55	O
ATOM	17	CB	LYS	A	2	6.951	3.406	5.907	1.00	43.90	C
ATOM	18	CG	LYS	A	2	6.088	3.566	4.640	1.00	50.06	C
ATOM	19	CD	LYS	A	2	6.084	4.892	3.921	1.00	54.79	C
ATOM	20	CE	LYS	A	2	7.505	5.186	3.441	1.00	59.95	C
ATOM	21	NZ	LYS	A	2	7.488	6.172	2.371	1.00	65.93	N

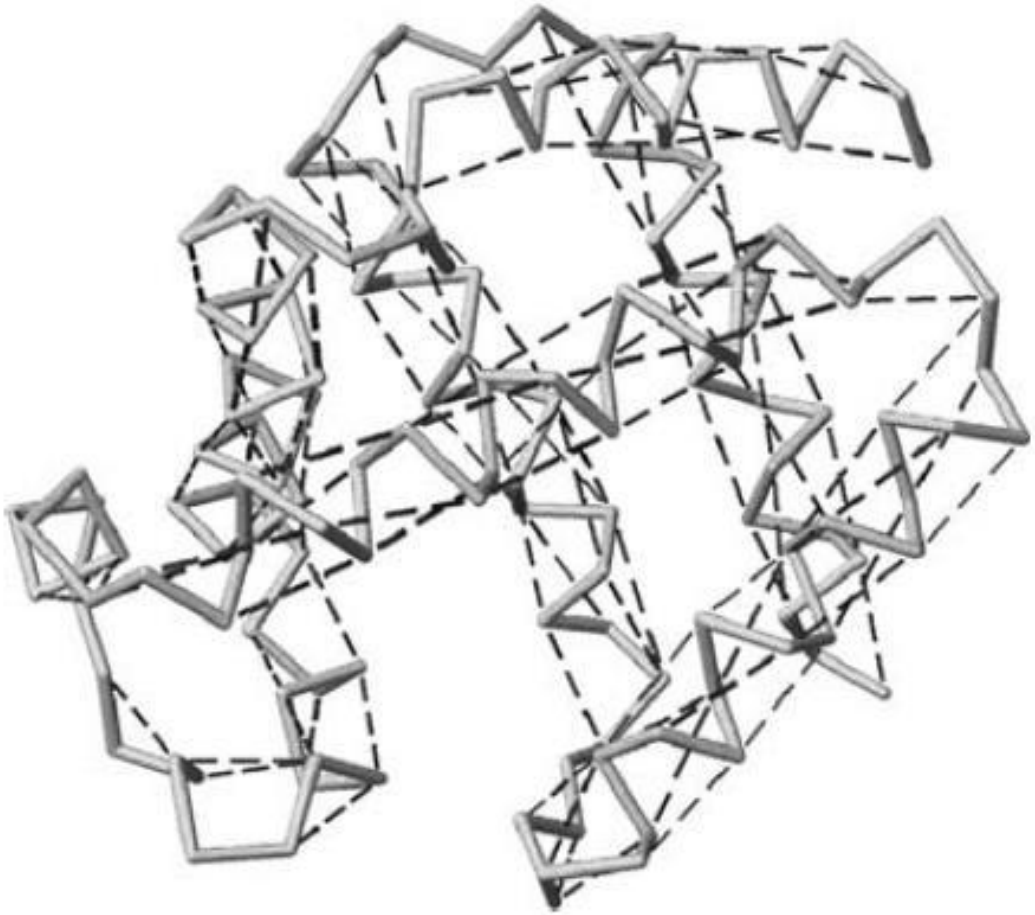


Fig. 14.5

Hydrogen bonds in a protein.
Hydrogen bonds (dotted lines) between amino acids in the protein represented in Figure 14.2.

Noncovalent interactions in RIG model

Node: Amino acid
Link: Noncovalent interactions

Noncovalent interactions gathered via spatial proximity capture electrostatic, hydrophobic, van der Waals forces, ionic interactions, hydrogen bonds etc.

- Non-covalent interactions are very weak in comparison with covalent bonds.
- They decay very quickly with the distance separating the interacting atoms. For instance, the energy of charge–charge interactions decays with $^{1}/_{r}$, where r is the distance separating the interacting species.
- However, dispersion and van der Waals forces decay with $^{1}/_{r^6}$ and $^{1}/_{r^{12}}$, respectively.
- Therefore, although the distance at which a pair of residues in a protein interact changes according to the type of interaction taken into account, it is almost certain that such interactions do not exist for residues separated at distances larger than 10 Å.

“Distance” between two amino acids

An amino acid (residue) is formed by several atoms, and we can define the distance between both residues in several different ways.

Different measures of inter-residue distances:

- **Distances between C_α atoms of two amino acids**: This measure does not take into account the proximity effects between atoms in the side chains of the residues.
- **Distances between the centres of residues**. This measure takes into account the proximity effects between side chains, but is more time-consuming than those based on C_α distance.
- **Distances between C_β of the residues**: This measure attempts to transform the information contained in the 3-D structure of a protein into a residue graph.
- **Distances between centroids of residues**: Instead of relying on a specific atom in the residue, this measure identifies centroid of the residue, which is used as ts positional representative.

“Distance” between two amino acids

The distance between two residues i and j represented by the Cartesian distance between them,

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

where x_i , y , z_i are the coordinates for the C_α / C_β atom of the residue i .

ATOM	1	N	ALA	A	0	11.081	1.847	9.557	1.00	64.40	N
ATOM	2	CA	ALA	A	0	10.369	0.997	10.519	1.00	60.28	C
ATOM	3	C	ALA	A	0	9.181	0.318	9.833	1.00	56.39	C
ATOM	4	O	ALA	A	0	9.344	-0.374	8.829	1.00	52.67	O
ATOM	5	N	ASN	A	1	7.992	0.669	10.335	1.00	55.08	N
ATOM	6	CA	ASN	A	1	6.691	0.239	9.830	1.00	53.00	C
ATOM	7	C	ASN	A	1	6.406	0.688	8.391	1.00	47.79	C
ATOM	8	O	ASN	A	1	5.757	-0.071	7.678	1.00	46.55	O
ATOM	9	CB	ASN	A	1	5.615	0.752	10.808	1.00	60.06	C
ATOM	10	CG	ASN	A	1	4.159	0.416	10.450	1.00	65.81	C
ATOM	11	OD1	ASN	A	1	3.751	-0.731	10.250	1.00	68.33	O
ATOM	12	ND2	ASN	A	1	3.303	1.433	10.367	1.00	68.59	N
ATOM	13	N	LYS	A	2	6.837	1.833	7.838	1.00	43.12	N
ATOM	14	CA	LYS	A	2	6.677	1.983	6.389	1.00	38.31	C
ATOM	15	C	LYS	A	2	7.612	1.040	5.632	1.00	31.24	C
ATOM	16	O	LYS	A	2	7.173	0.454	4.658	1.00	35.55	O
ATOM	17	CB	LYS	A	2	6.951	3.406	5.907	1.00	43.90	C
ATOM	18	CG	LYS	A	2	6.088	3.566	4.640	1.00	50.06	C
ATOM	19	CD	LYS	A	2	6.084	4.892	3.921	1.00	54.79	C
ATOM	20	CE	LYS	A	2	7.505	5.186	3.441	1.00	59.95	C
ATOM	21	NZ	LYS	A	2	7.488	6.172	2.371	1.00	65.93	N

Residue Interaction Graph

$$A_{ij} = \begin{cases} H\left(r_C - r_{ij}\right) & i \neq j \\ 0 & i = j, \end{cases}$$

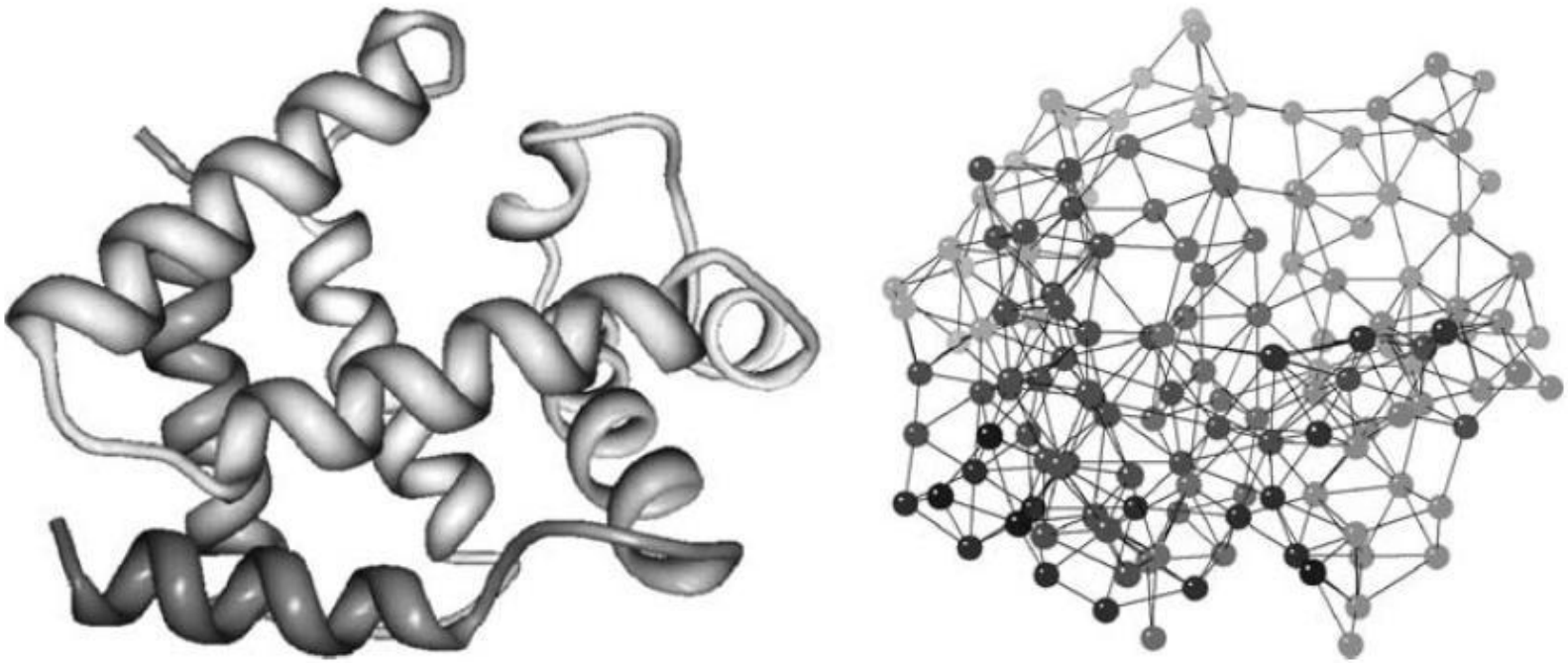
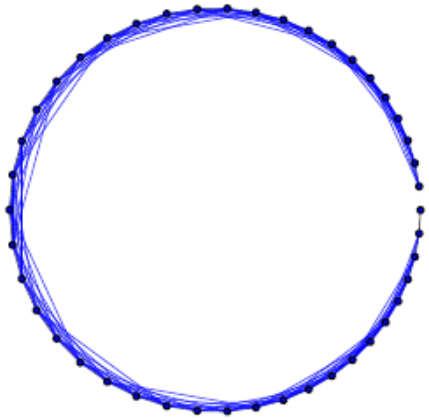
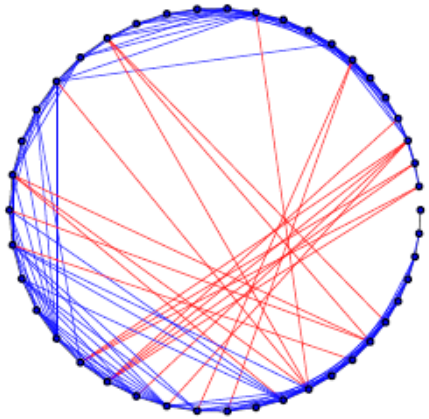


Fig. 14.6
Protein residue network. A protein residue network (right) obtained for the protein with PDB code 1ASH (left).

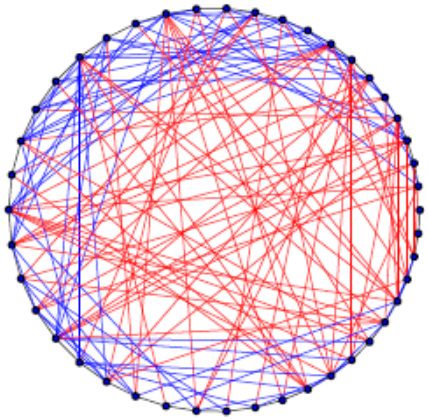
Small-World Nature of RIGs



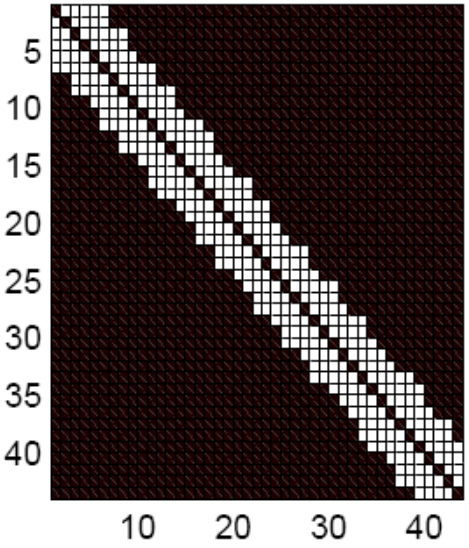
REGULAR CONTROL

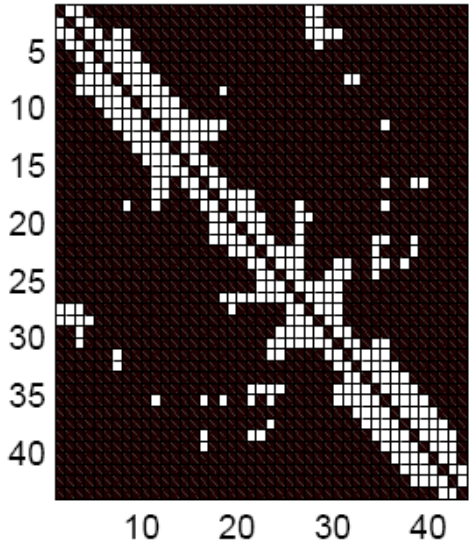


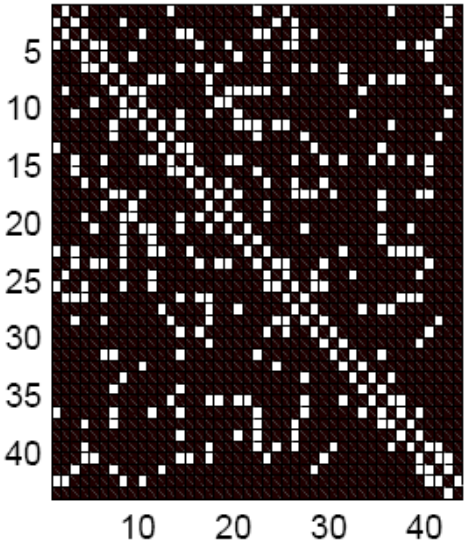
PCN



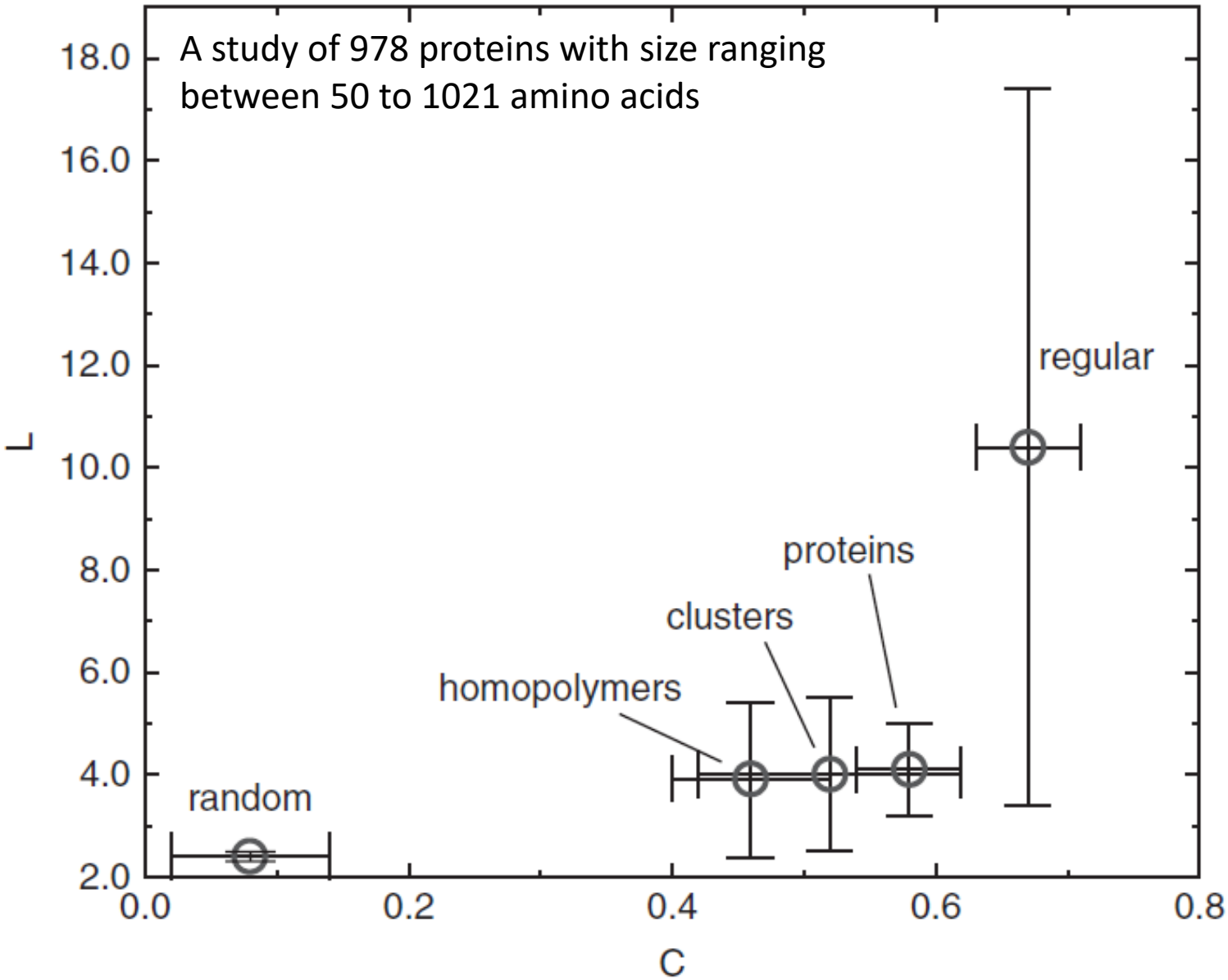
RANDOM CONTROL







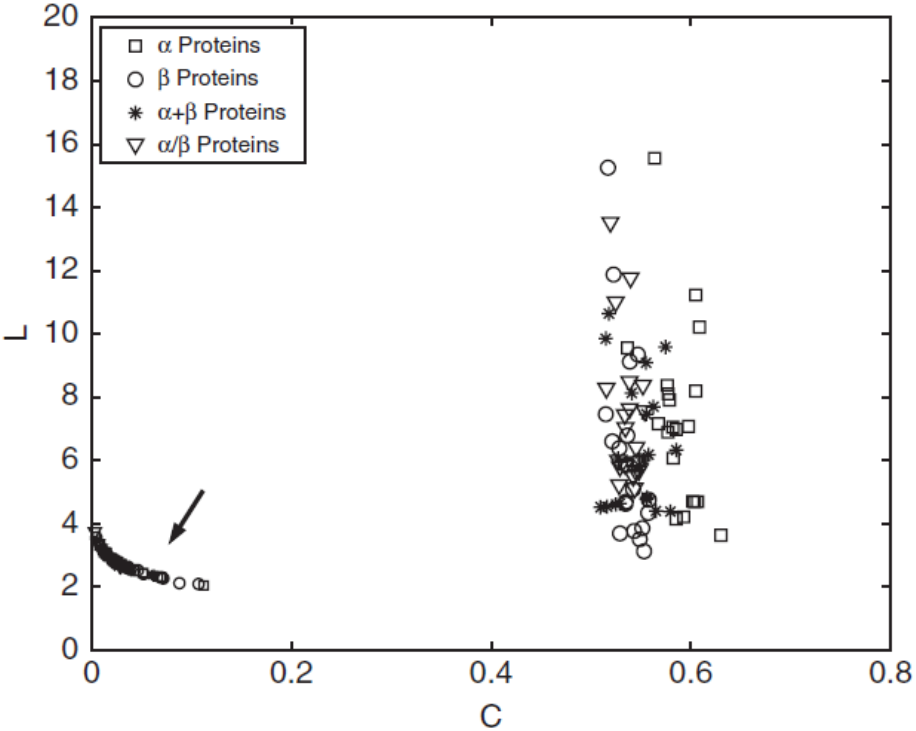
Small-World Nature of RIGs

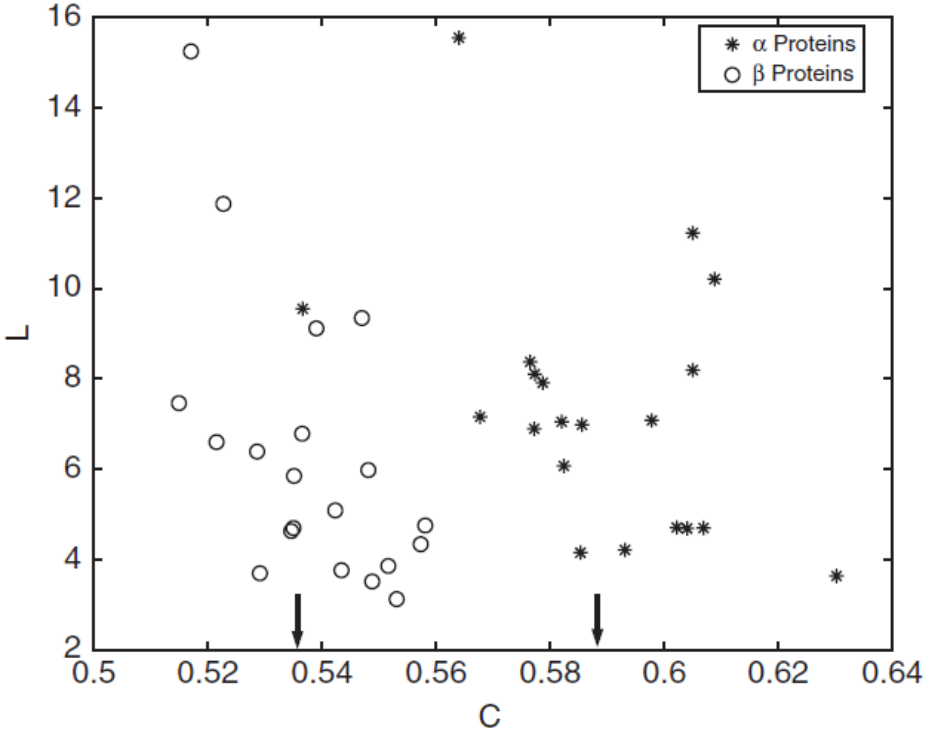


Vendruscolo *et al.*, Small-world view of the amino acids that play a key role in protein folding. *Phys. Rev. E* **65**, 061910 (2002).

Small-World Nature of RIGs

A study of 80 proteins with a range of 73 to 2,359 amino acids, classified into four structural classes according to their secondary structures: *α, β, α + β* and *α/β*.

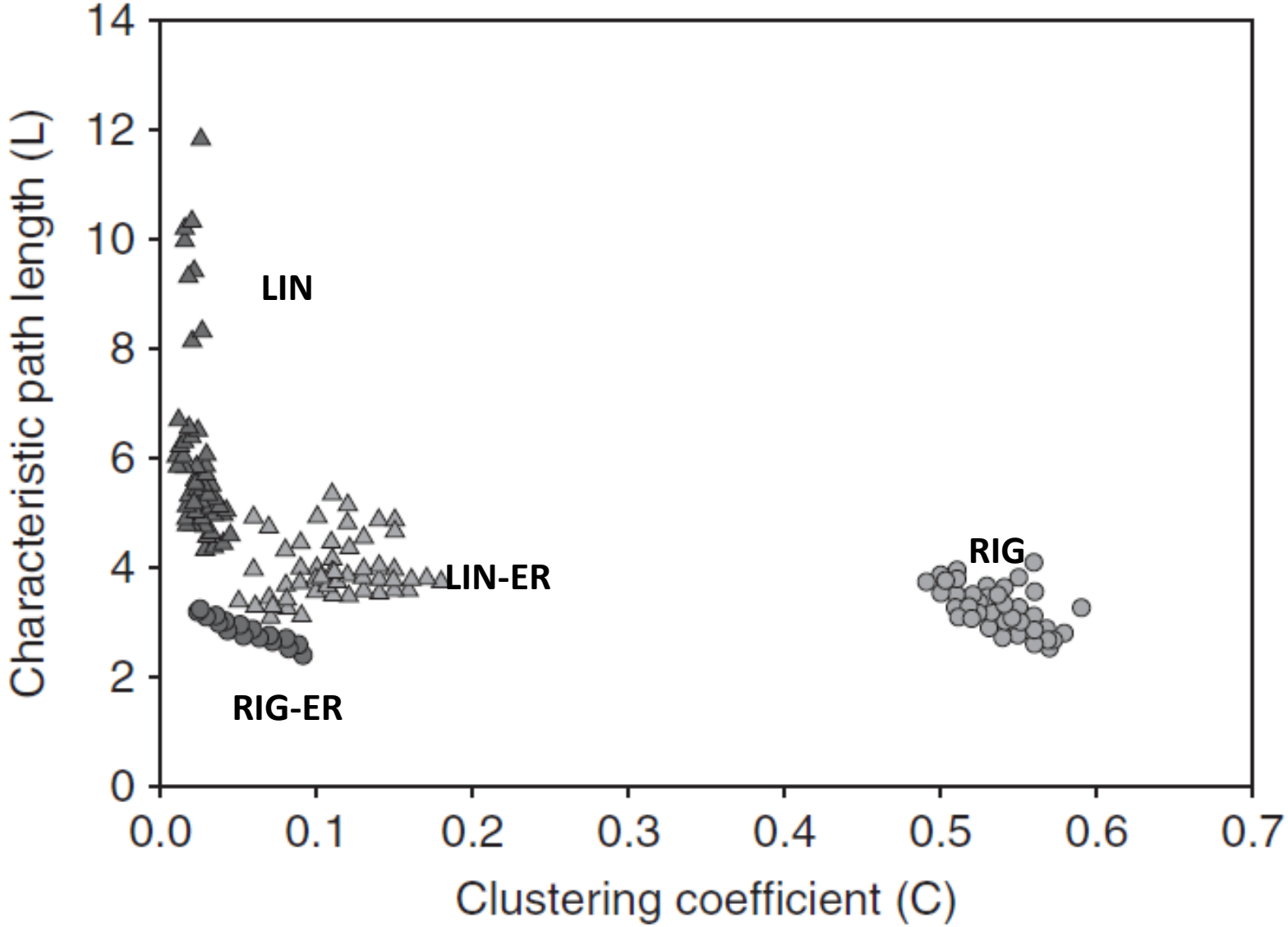




The *α* and *β* proteins are those having mainly helices and sheets in their structures, respectively. *α + β* and *α/β* proteins have combinations of both types of secondary structures.

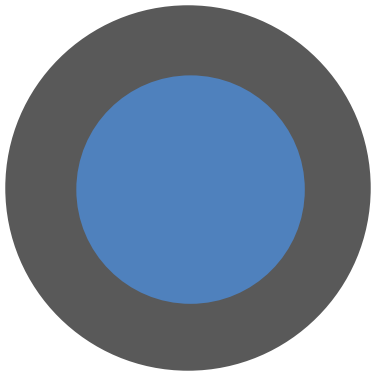
Bagler and Sinha, Network properties of protein structures. *Physica A* **346**, 27–33 (2005).

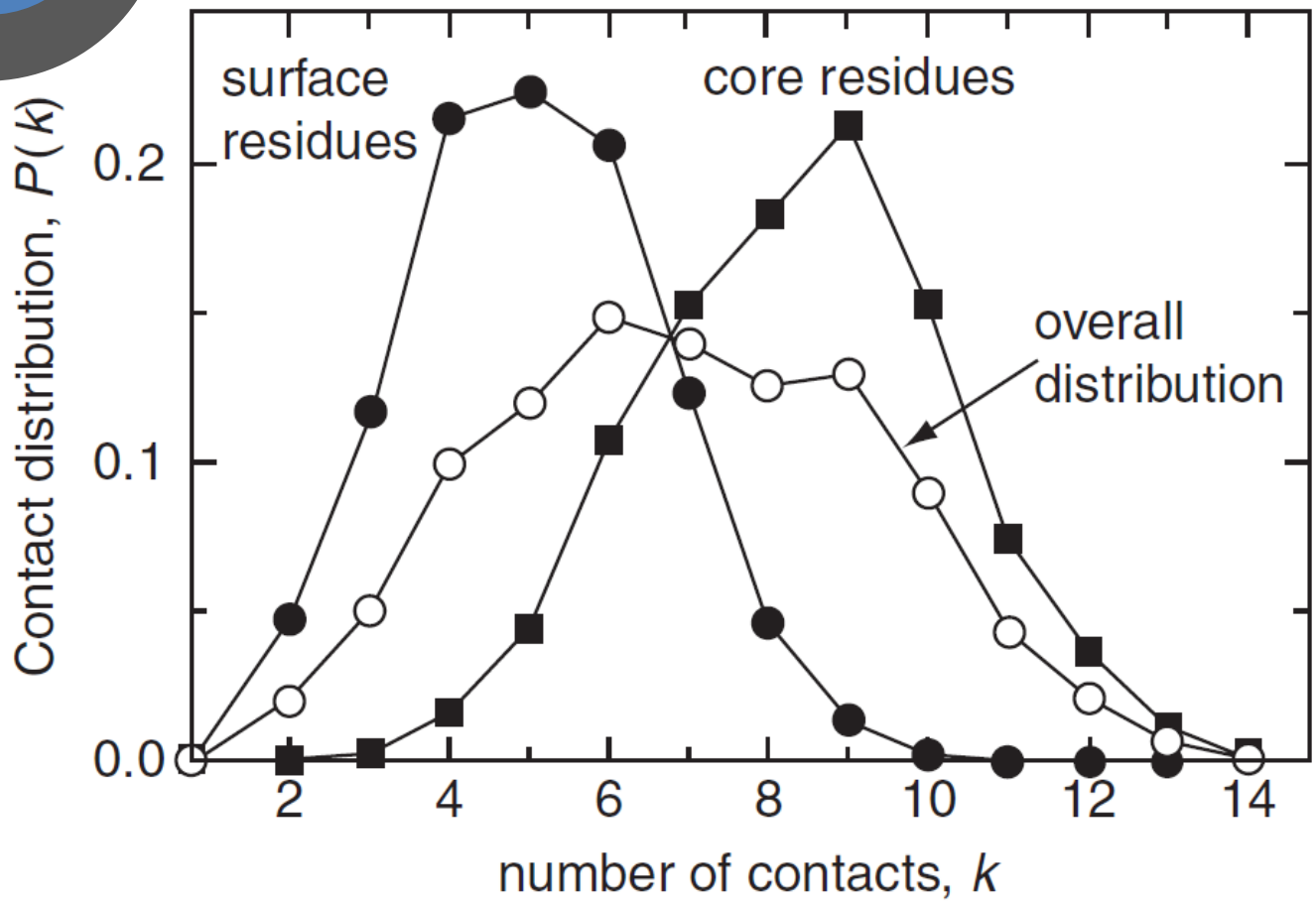
Small-World Nature of RIGs and LINs



Greene and Higman, Uncovering network systems within protein structures, *J. Mol. Biol*. **334**, 781–91 (2003).

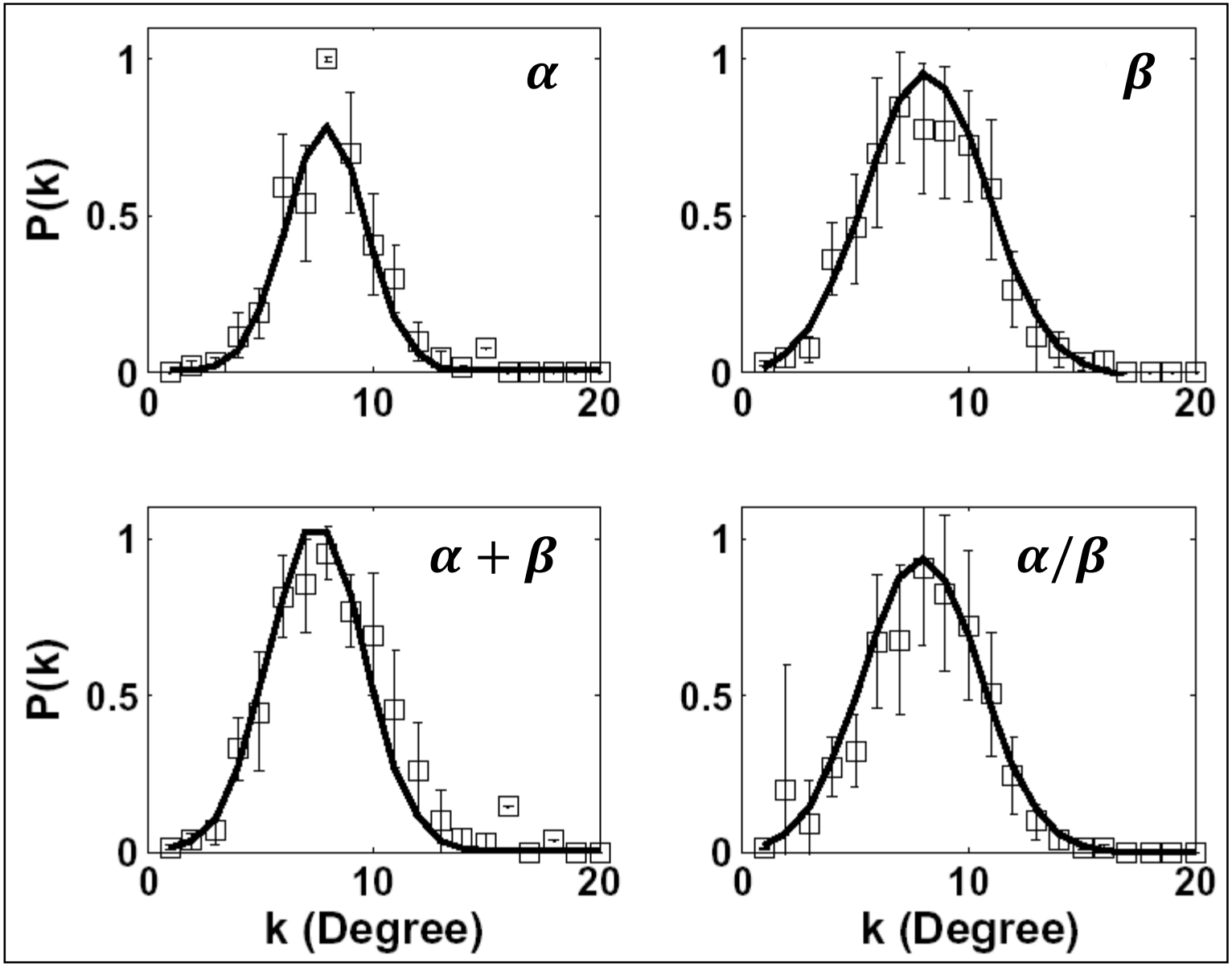
Degree distribution of RIGs: Separating core and surface





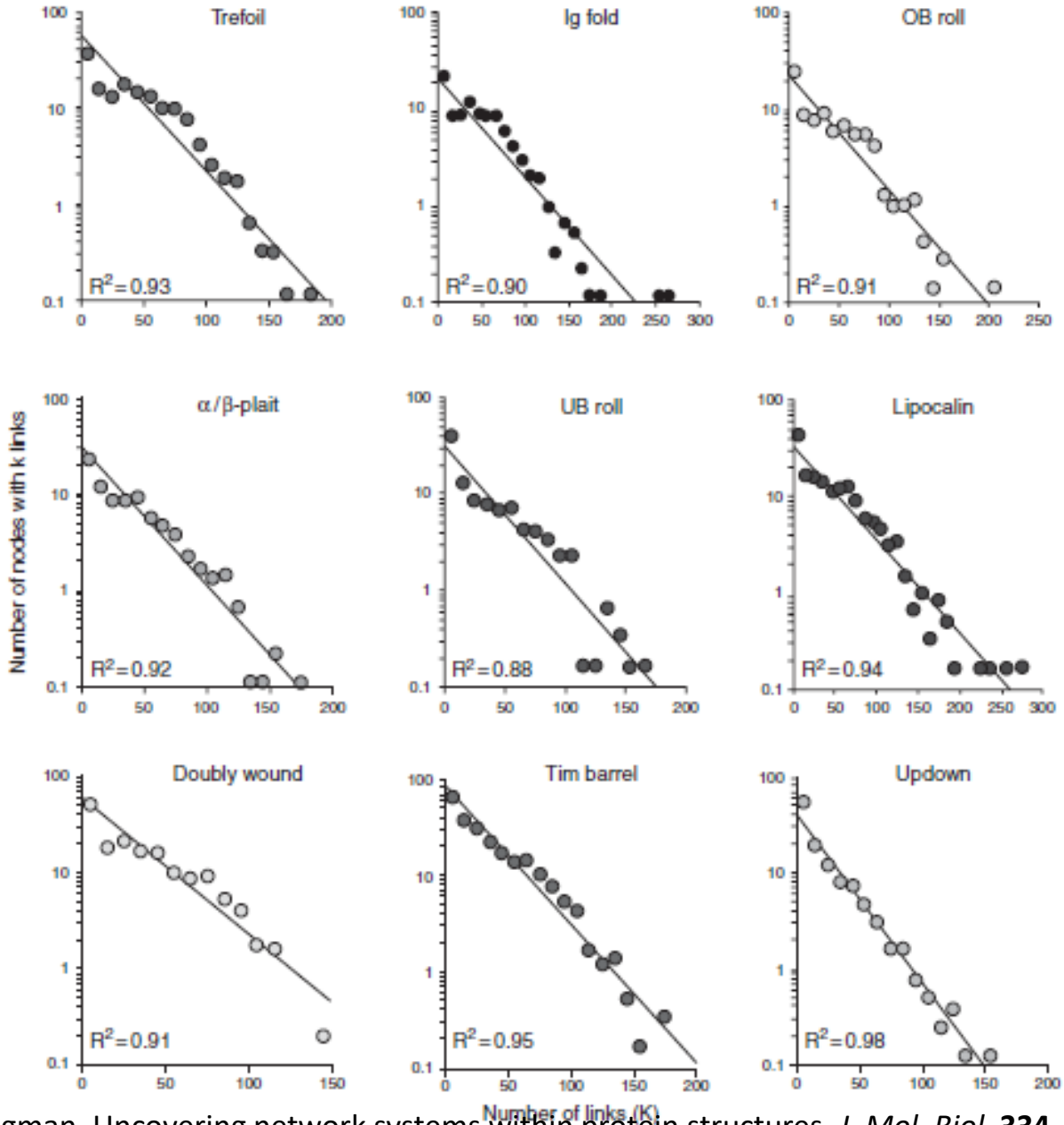
Atilgan *et al.* Small-world communication of residues and significance for protein dynamics, *J. Biophys. J.* **86**, 85–91 (2004)

Degree distribution of RIGs: Across structural classes



Bagler and Sinha, Network properties of protein structures. *Physica A* **346**, 27–33 (2005).

Degree distribution of RIGs: Across folds/domains



Greene and Higman, Uncovering network systems within protein structures, *J. Mol. Biol*. **334**, 781–91 (2003).

Degree distribution of RIGs: Segregating contacts of amino acids

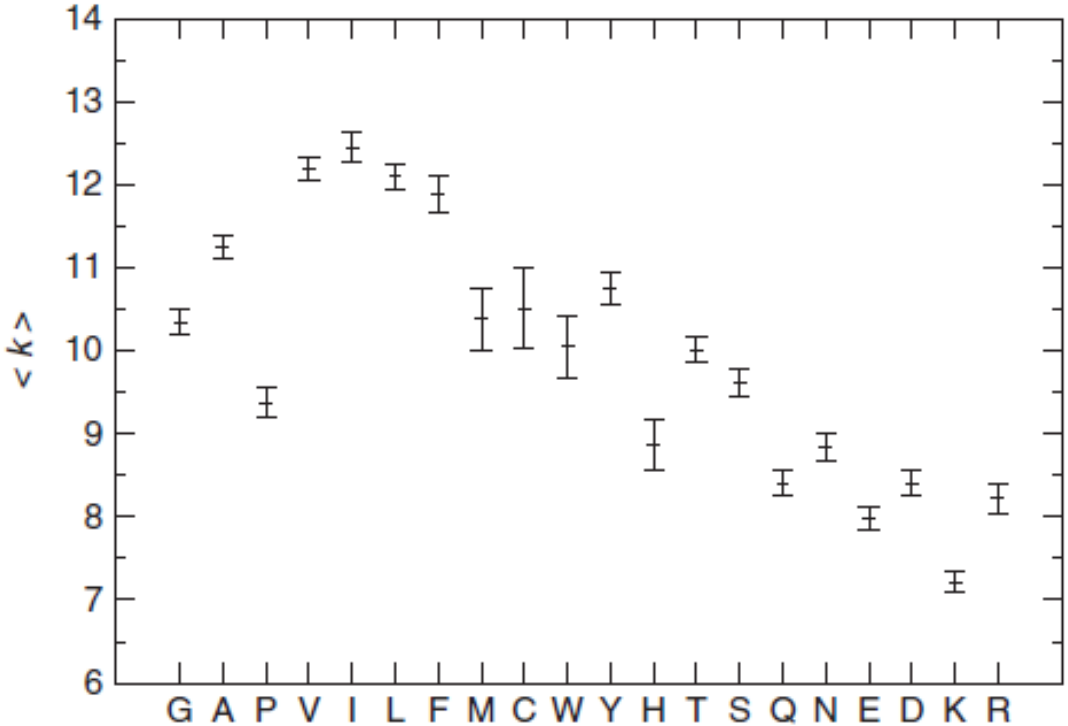
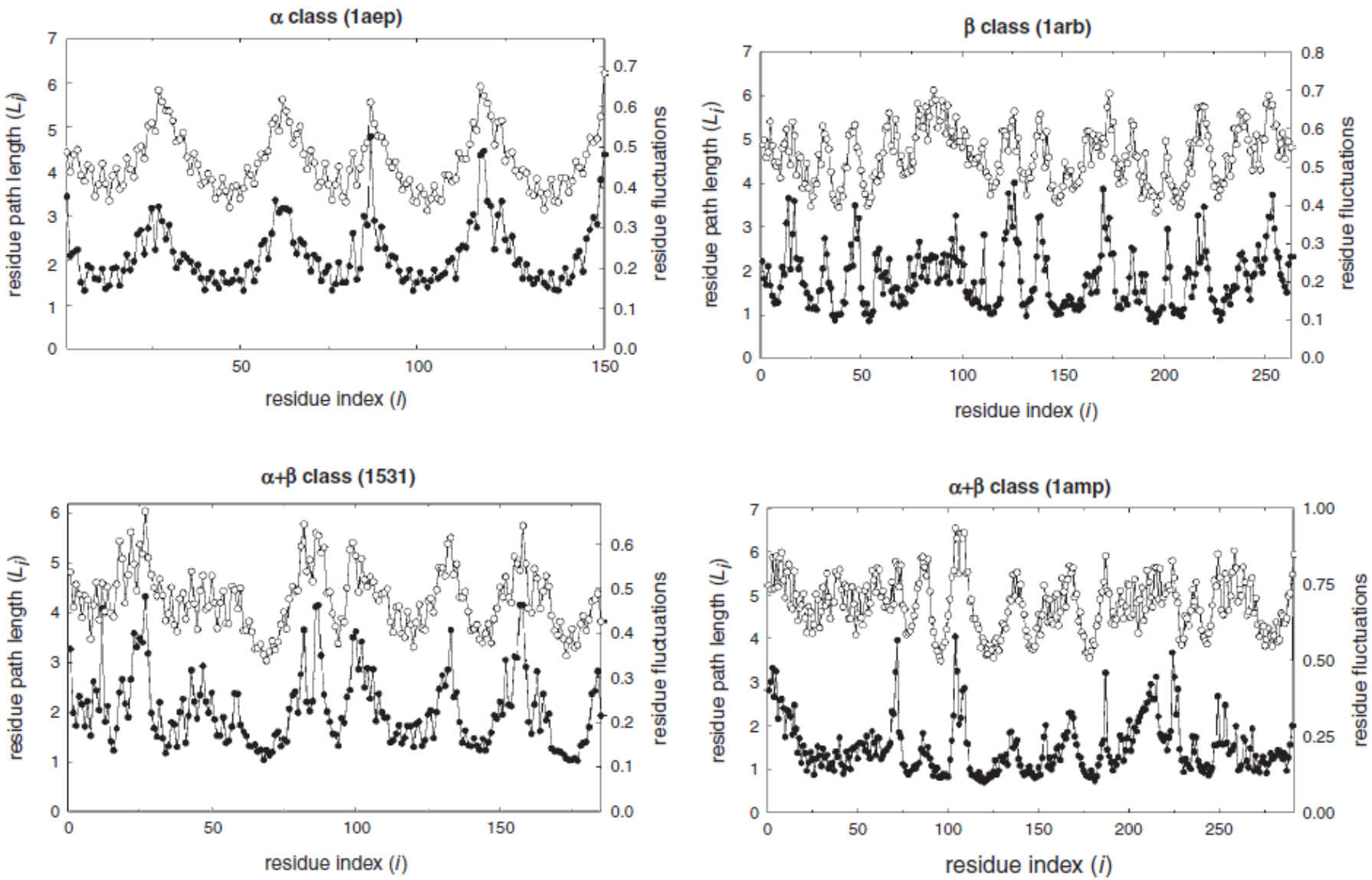


Fig. 14.14
Average degree of amino acids in proteins. Representation of the average degree and their standard deviations for each of the twenty natural amino acids occurring in proteins. From Alves, N. A., and Martinez, A. S. (2007). Inferring topological features of proteins from amino acid residue networks. *Physica A* **375**, 336–344. (Reproduced with the permission of Elsevier, ©2007.)

Closeness as a proxy for thermal fluctuations



BIOINFORMATICS ORIGINAL PAPER Vol. 23 no. 14 2007, pages 1760–1767
doi:10.1093/bioinformatics/btm257

Structural bioinformatics

Assortative mixing in Protein Contact Networks and protein folding kinetics

Ganesh Bagler and Somdatta Sinha*

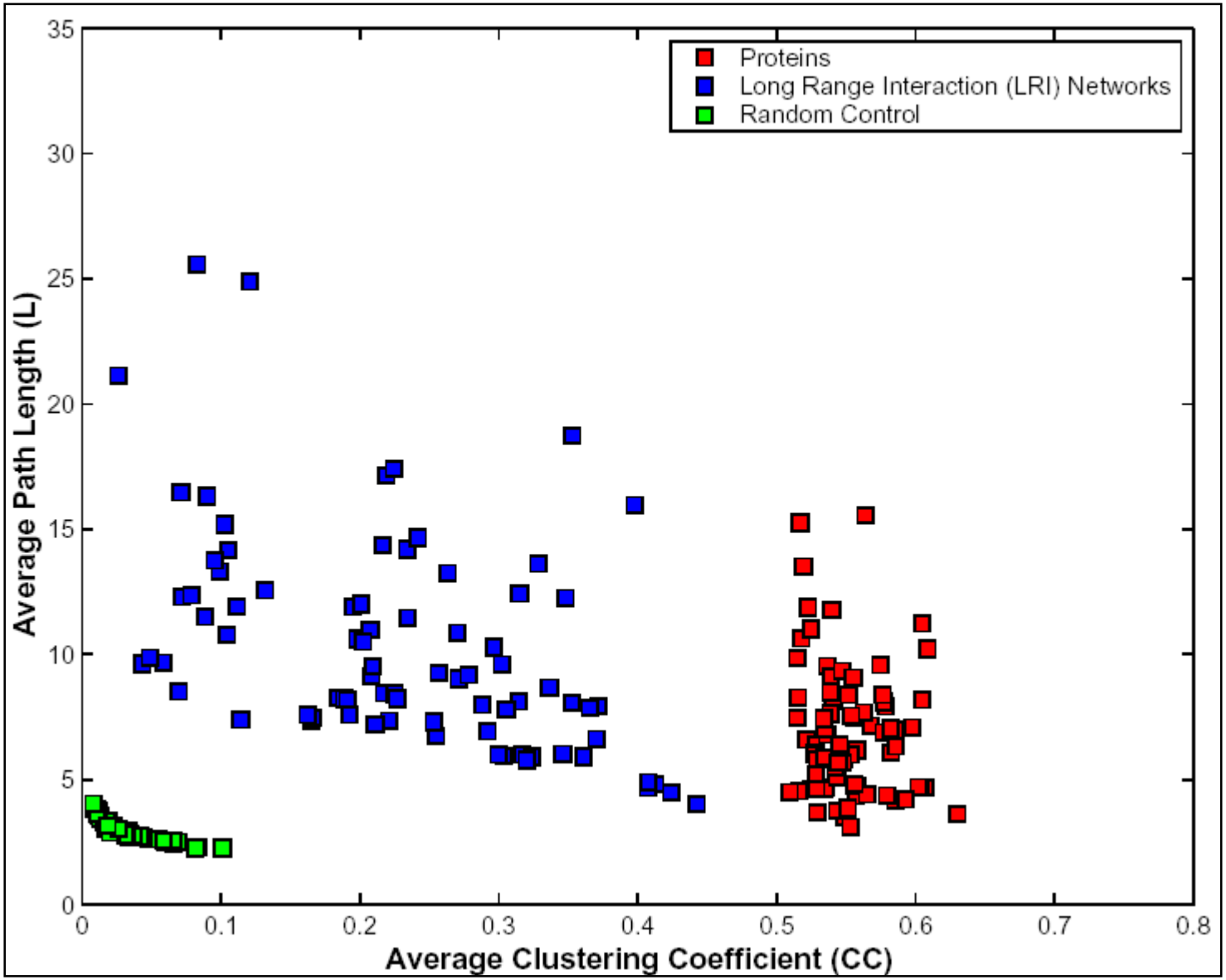
Centre for Cellular and Molecular Biology, Uppal Road, Hyderabad 500007, India

Received on February 5, 2007; revised on May 8, 2007; accepted on May 8, 2007

Advance Access publication May 22, 2007

Associate Editor: Alfonso Valencia

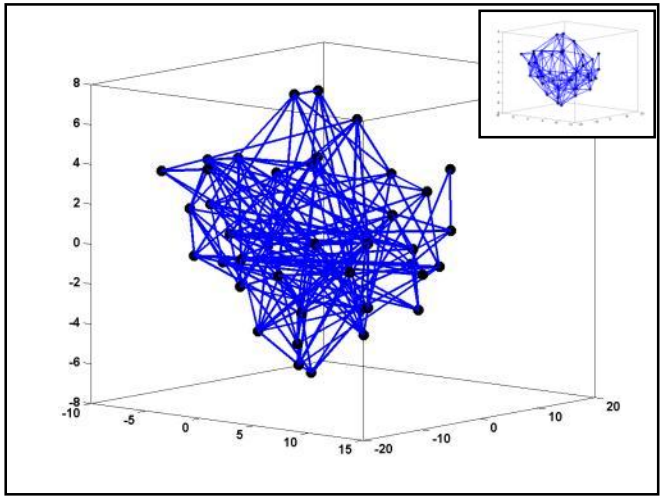
L—C PLOT FOR 80 PROTEINS

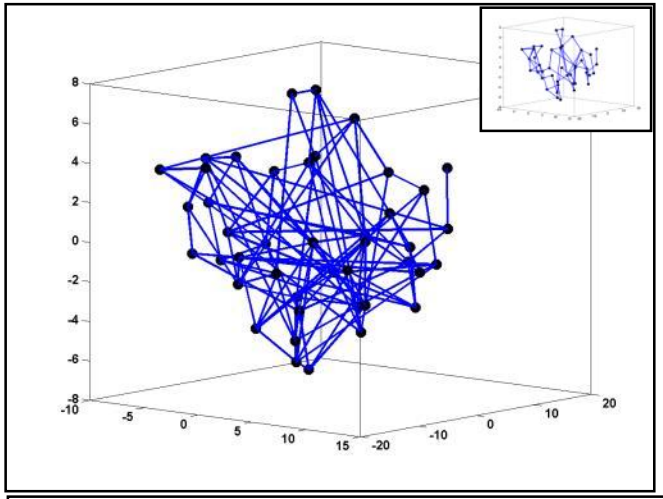


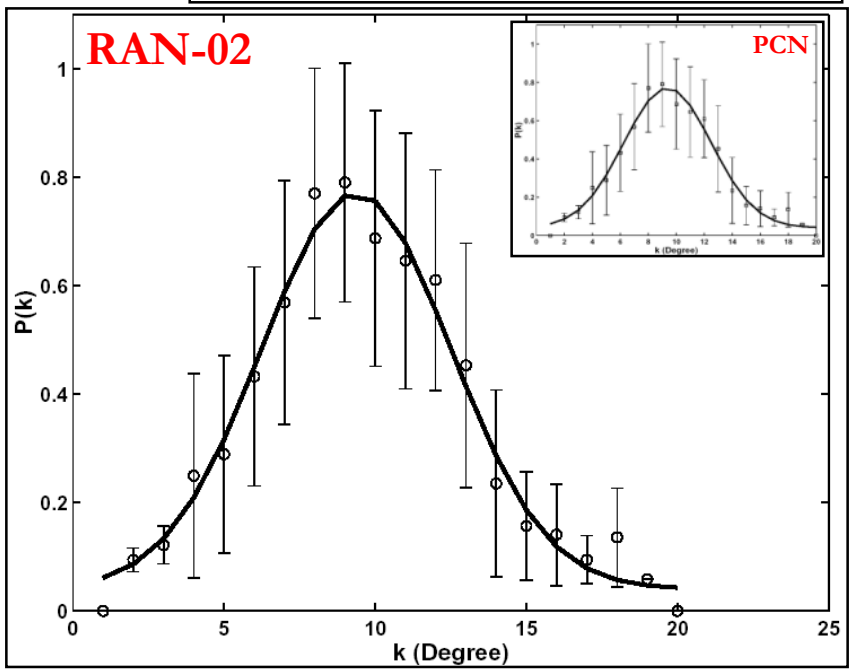
Bagler and Sinha, Bioinformatics (2007)

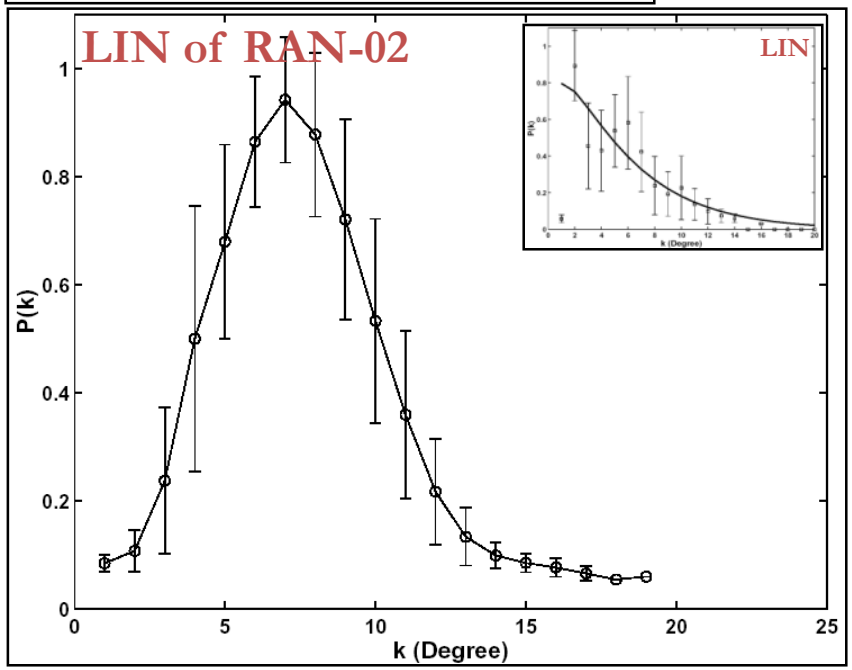
DEGREE DISTRIBUTION OF A NETWORK

(TYPE II) Random Controls & their LINs





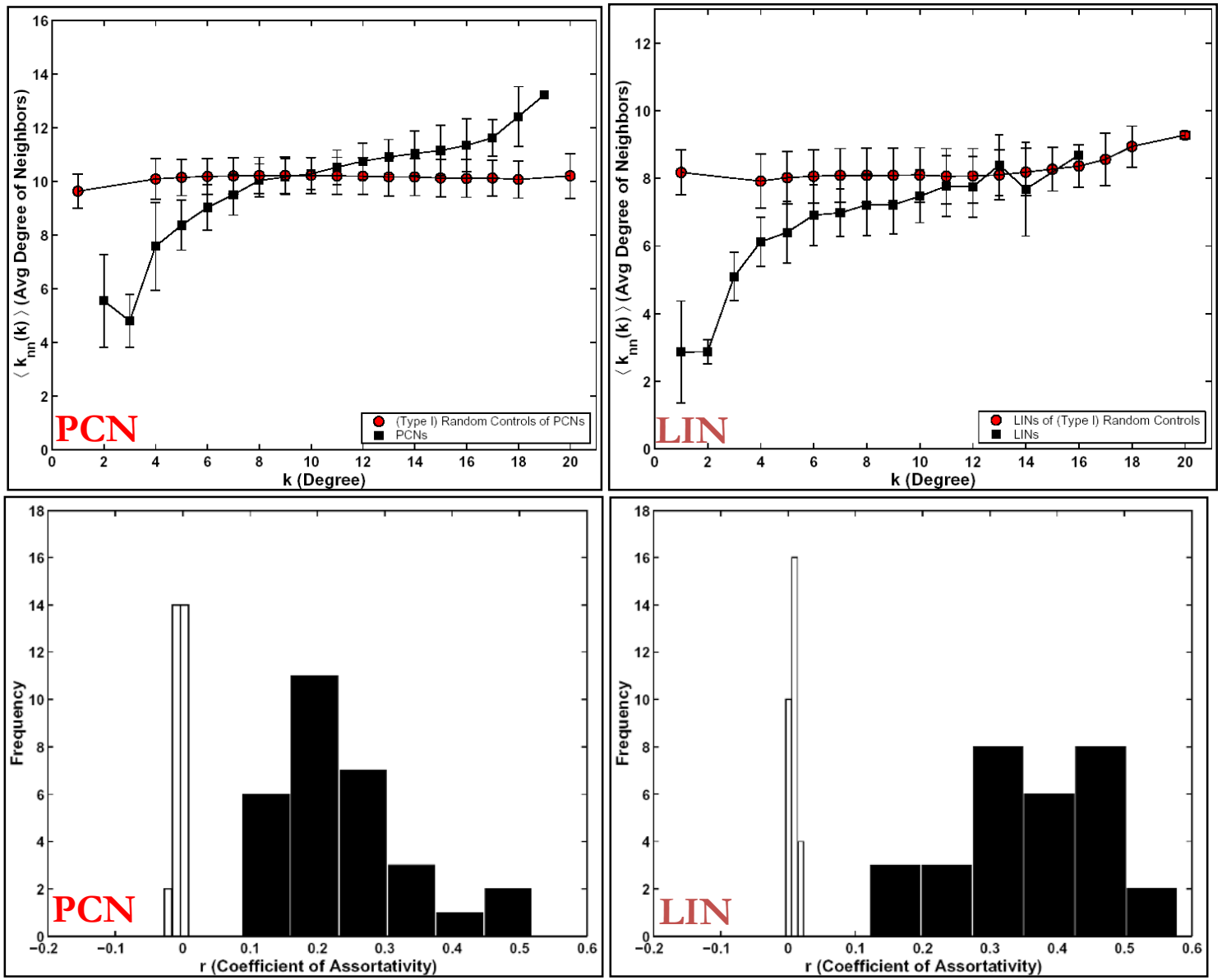




Ganesh Bagler and Somdatta Sinha, Bioinformatics, (2007).

ASSORTATIVE DEGREE MIXING IS PARTIALLY EXPLAINED BY "DEGREE DISTRIBUTION"

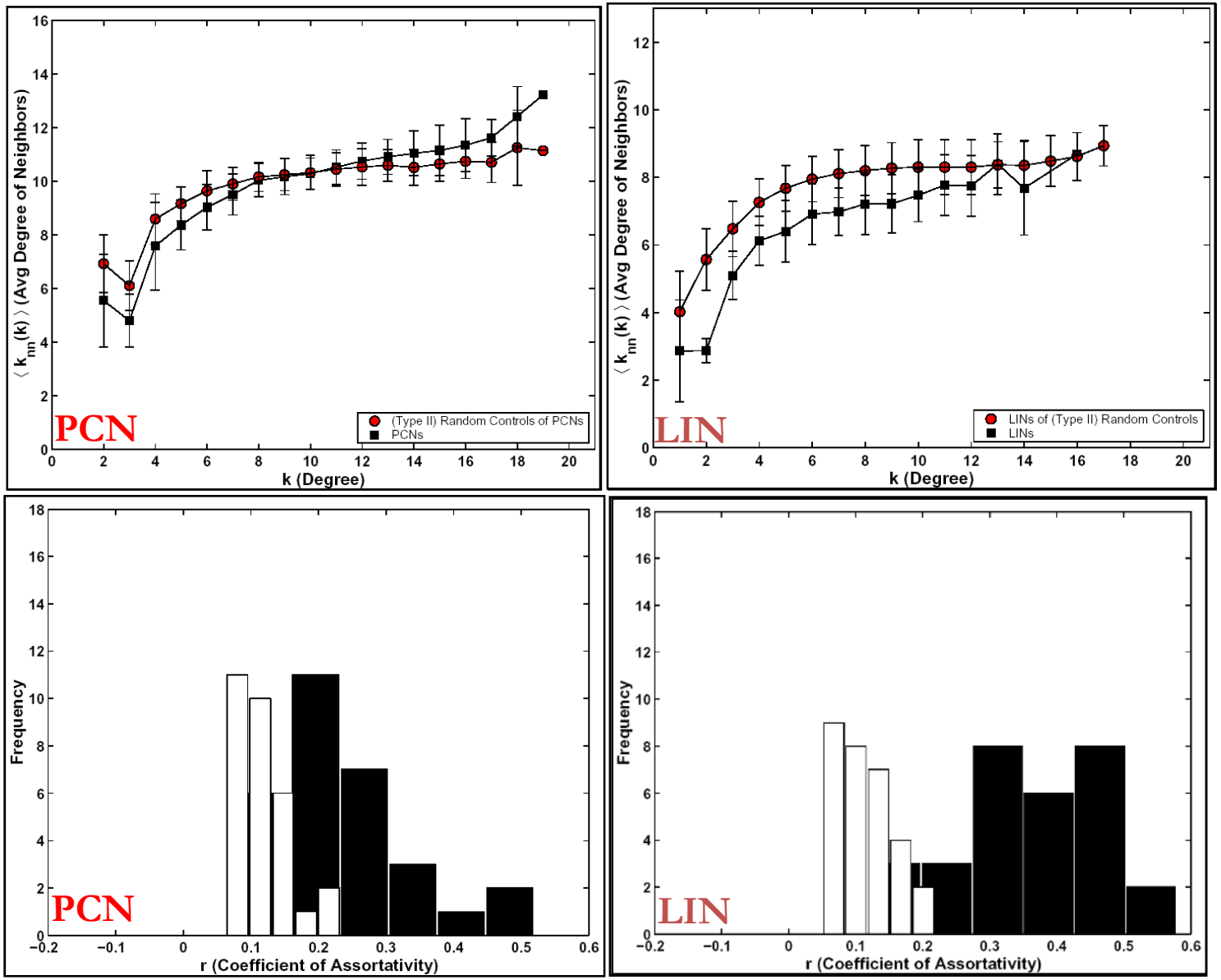
Ganesh Bagler and Somdatta Sinha, Bioinformatics, (2007).



PCNs and LINs have a clear signature of assortative mixing as compared to the Type-I Random Controls.

ASSORTATIVE DEGREE MIXING IS PARTIALLY EXPLAINED BY "DEGREE DISTRIBUTION"

Ganesh Bagler and Somdatta Sinha, Bioinformatics, (2007).



The observed assortativity, in both PCNs and LINs, is, partially, recovered in Type-II Random Controls.

RATE OF FOLDING STATS

	PDB_ID	ln(k_F)	Time of Folding
α	1hrc	8.76	0.1569 ms
	1imq	7.31	0.6688 ms
	1ycc	9.62	66.39 us
	2abd	6.55	1.43 ms
	2pdd	9.8	0.5545 us
αβ	1aps	-1.48	4.393 sec
	1cis	3.87	20.86 ms
	1coa	3.87	20.86 ms
	1fkb	1.46	0.2322 sec
	1hdn	2.7	67.21 ms
	1pba	6.8	1.114 ms
	1ubq	7.33	0.6556 ms
	1urn	5.76	3.1511 ms
	1vik	6.8	1.1134 ms
	2hqi	0.18	0.8352 sec
	2ptl	4.1	16.57 ms
	2vik	6.8	1.114 ms
β	1aey	2.09	0.1237 sec
	1csp	6.98	0.9303 ms
	1mjc	5.24	5.3 ms
	1nyf	4.54	10.67 ms
	1pks	-1.05	2.858 sec
	1shf	4.55	10.57 ms
	1shg	1.41	0.2441 sec
	1srl	4.04	17.6 ms
	1ten	1.06	0.3465 sec
	1tit	3.47	31.11 ms
	1wit	0.41	0.6637 sec
	2ait	4.2	15 ms
	3mef	5.3	4.992 ms

PDB_ID	ln(k_F)	Time of Folding
2pdd	9.8	0.5545 us
1ycc	9.62	66.39 us
1hrc	8.76	0.1569 ms
1ubq	7.33	0.6556 ms
1imq	7.31	0.6688 ms
1csp	6.98	0.9303 ms
1pba	6.8	1.114 ms
1vik	6.8	1.1134 ms
2vik	6.8	1.114 ms
2abd	6.55	1.43 ms
1urn	5.76	3.1511 ms
3mef	5.3	4.992 ms
1mjc	5.24	5.3 ms
1shf	4.55	10.57 ms
1nyf	4.54	10.67 ms
2ait	4.2	15 ms
2ptl	4.1	16.57 ms
1srl	4.04	17.6 ms
1cis	3.87	20.86 ms
1coa	3.87	20.86 ms
1tit	3.47	31.11 ms
1hdn	2.7	67.21 ms
1aey	2.09	0.1237 sec
1fkb	1.46	0.2322 sec
1shg	1.41	0.2441 sec
1ten	1.06	0.3465 sec
1wit	0.41	0.6637 sec
2hqi	0.18	0.8352 sec
1pks	-1.05	2.858 sec
1aps	-1.48	4.393 sec

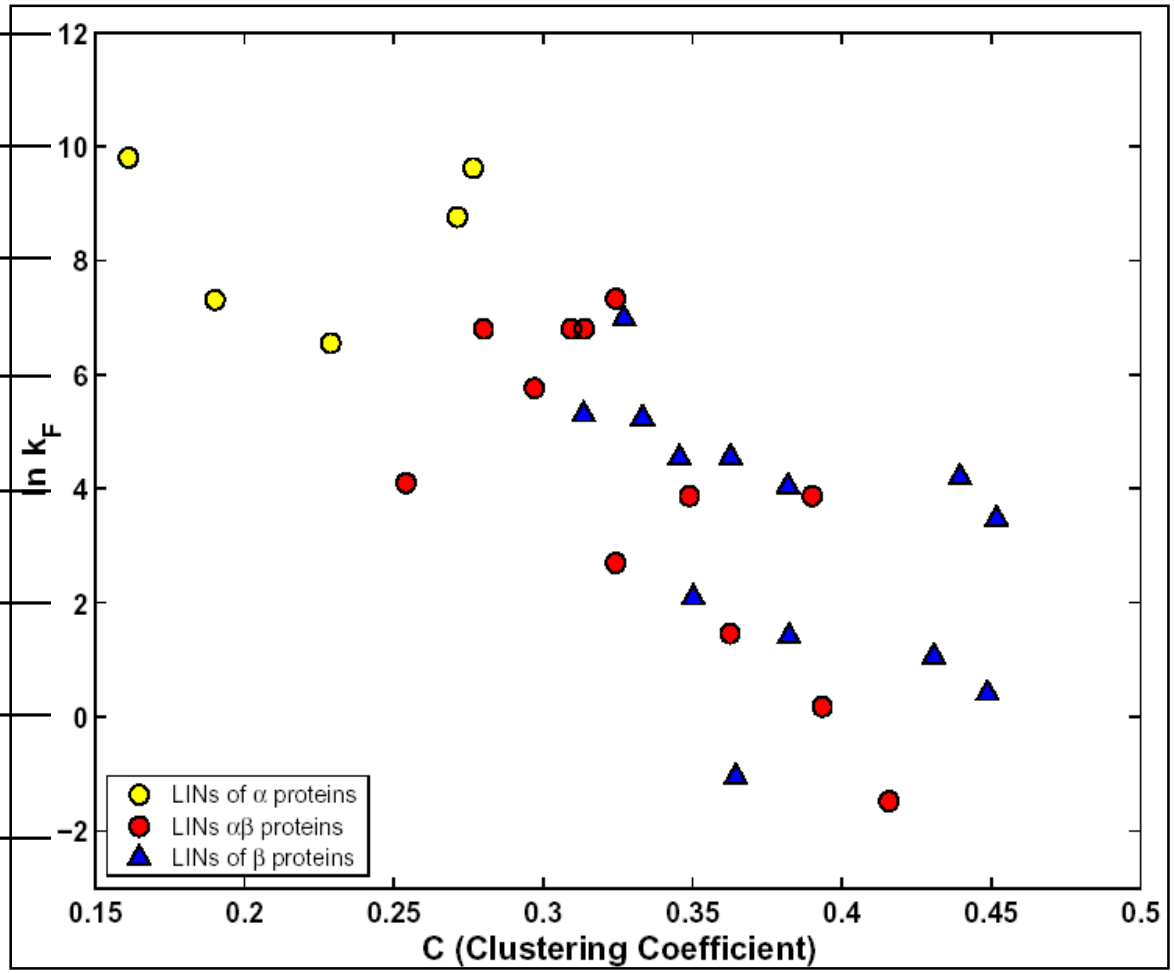
Bagler and Sinha, Bioinformatics (2007)

PDB_ID	ln(k_F)	Name	Functional Class
1hrc	8.76	HORSE HEART CYTOCHROME C	Electron Transport(cytochrome)
1imq	7.31	COLICIN E9 IMMUNITY PROTEIN IM9	Bacteriocin
1ycc	9.62	YEAST ISO-1-CYTOCHROME C	Electron Transport (cytochrome)
2abd	6.55	ACYL-COENZYME A BINDING PROTEIN FROM BOVINE LIVER	Acyl Coenzyme a Binding Protein
2pdd	9.8	ACETYLTRANSFERASE	Oxido Reductase(acyltransferase)
1aps	-1.48	ACYLPHOSPHATASE	Hydrolase(acting On Acid Anhydrides)
1cis	3.87	CHYMOTRYPSIN INHIBITOR 2 AND HELIX E	Hybrid Protein
1coa	3.87	THE HYDROPHOBIC CORE OF CHYMOTRYPSIN INHIBITOR 2	Serine Protease Inhibitor
1fkb	1.46	RAPAMYCIN HUMAN IMMUNOPHILIN FKBP-12 COMPLEX	Isomerase
1hdn	2.7	PHOSPHOCARRIER PROTEIN HPR FROM E. COLI	Phosphotransferase
1pba	6.8	ACTIVATION DOMAIN OF PORCINE PROCARBOXYPEPTIDASE B	Hydrolase(c Terminal Peptidase)
1ubq	7.33	UBIQUITIN	Chromosomal Protein
1urn	5.76	U1A MUTANT/RNA COMPLEX + GLYCEROL	Transcription/rna
1vik	6.8	HIV-1 PROTEASE	Aspartyl Protease
2hqi	0.18	OXIDIZED FORM OF MERP	Transport
2ptl	4.1	IMMUNOGLOBULIN LIGHT CHAIN-BINDING DOMAIN OF PROTEIN L	Binding Protein
2vik	6.8	ACTIN-SEVERING DOMAIN VILLIN 14T	Actin Binding Protein
1aey	2.09	ALPHA-SPECTRIN SRC HOMOLOGY 3 DOMAIN	Cytoskeleton
1csp	6.98	BACILLUS SUBTILIS MAJOR COLD SHOCK PROTEIN	Transcription Regulation
1mjc	5.24	THE MAJOR COLD SHOCK PROTEIN OF ESCHERICHIA COLI	Transcription Regulation
1nyf	4.54	SH3 DOMAIN FROM FYN PROTO-ONCOGENE TYROSINE KINASE	Phosphotransferase
1pks	-1.05	THE PI3K SH3 DOMAIN	Phosphotransferase
1shf	4.55	THE SH3 DOMAIN IN HUMAN FYN	Phosphotransferase
1shg	1.41	SRC-HOMOLOGY 3 (SH3) DOMAIN	Cytoskeleton
1srl	4.04	THE SRC SH3 DOMAIN	Phosphotransferase
1ten	1.06	FIBRONECTIN TYPE III DOMAIN FROM TENASCIN	Cell Adhesion Protein
1tit	3.47	TITIN, IG REPEAT 27	Immunoglobulin Like Domain
1wit	0.41	TWITCHIN IMMUNOGLOBULIN SUPERFAMILY DOMAIN	Muscle Protein
2ait	4.2	ALPHA-AMYLASE INHIBITOR TENDAMISTAT	Alpha Amylase Inhibitor
3mef	5.3	MAJOR COLD-SHOCK PROTEIN FROM ESCHERICHIA COLI	Gene Regulation

RATE OF FOLDING—FOLDING TIME

ln (k_F)	k_F [sec^{-1}]	Folding Time
12	16275.47	6.144 µs
10	22,026.00	0.4540 µs
8	2,981.00	33.55 ms
6	403.43	2.479 ms
4	54.60	18.32 ms
2	7.39	0.1353 s
0	1.00	1 s
-2	0.1353	7.389 s

Bagler and Sinha, Bioinformatics (2007)



Engineering a thermostable enzyme

SCIENTIFIC REPORTS
nature publishing group npg

"*Engineering a thermo-stable superoxide dismutase functional at sub-zero to >50°C, which also tolerates autoclaving*", Arun Kumar *et al.*, Scientific Reports (Nature Publishing Group), 2 (387), 1-8 (2012).

A simple model for generating contact maps

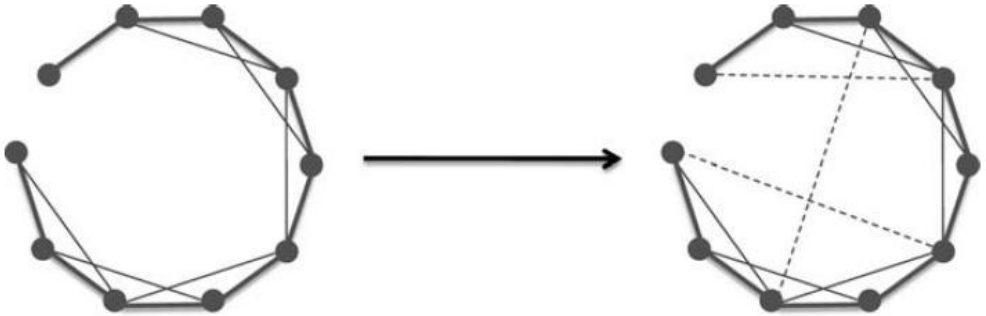
Procedure for generating contact maps using Bartoli's model:

(i) Assign 1s to the first two diagonals (up and down the main diagonal) of the adjacency matrix in order to define the backbone contacts.

(ii) Randomly select a pair of residues *i* and *j* with a probability that decreases linearly with the distance separating these residues in the protein sequence.

(iii) Assign 1s to the entries of the adjacency matrix corresponding to all nine residue pairs generated by the Cartesian product of $\{i-1, i, i+1\} \times \{j-1, j, j,+1\}$.

(iv) Iterate the last procedure until the number of links in the random graph is *close to* those of the real protein.



- The characteristic path length and clustering coefficient are not useful quantities for "protein fingerprinting",' because **they can be reproduced by using random networks in which constraints similar to those induced by the backbone connectivity are imposed**.
- Perhaps that's why the networks generated by using only **long-range interactions** are indistinguishable from random graphs.